MATH 2394

Engineering Maths

Topic 3: Introduction to Limits and Differentiation

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Activity: What happened in that frame?

Activity: What happened in that frame?

Activity: What happened in that frame?

?

Activity: What happened in that frame?

Activity: Reflection

- What is the process to obtain the likely event in the missing frame?
- What are some real-life examples where we have to guess the “most likely” event that is going to happen or had happened?

Today's Topics

1. **The Limits of a Function:** does a limit exist and what are its implications?

2. **Derivatives and Rate of Change:** identifying the rate of change of a function at any given point.

3. **Differentiation Formulas:** The rules of taking derivatives and when to use them.

4. **Derivatives of Special Function:** rate of change of trigonometric, exponential, and inversed functions.

5. **Applications of Differentiations:** useful ways derivatives could be used in the real world.

The limit of a function

To infinity and beyond

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Example

- A bank is now offering a special saving account where the account's balance doubles (gain 100% value) once every 12 months. I decide to deposit $100.00 into this account → $\$100 * (1 + 1) = \$200$ once a year.
- Twice a year at 100%/2 = 50% interest rate → $\$100 * \left(1 + \frac{1}{2}\right) * \left(1 + \frac{1}{2}\right) = \$225$.
- Seeing that the total amount increased by $25, I am very happy so I want to collect interest every month at 100%/12 = 8.333% each month→ $\$100 * \left(1 + \frac{1}{12}\right)^{12} = \$261.3$

Example

- **<u>Question:</u>** can I keep doing this forever and ask to collect interest every second, or 1/10 of a second, to get infinite money? In other words, what is the result of → $\$100 * \left(1 + \frac{1}{n}\right)^n$ where n becomes extremely big?

- **<u>Answer:</u>** No, the most I can get back is $271.82 after a year. This is also the definition of the Euler's number e.
- **<u>This is called finding a limit</u>**.

Example

- This act of approximation for a quantity A which is dependent on another quantity B as B approaches another value C is called **“taking the limit of A(B) as B approaches C”.**

Limits of a function– Main topics

- Definition and notation of a limit
- Calculating limits
- Examples

Limits of a function – Definition and Notation

- Recall the previous example, we have a function:

$$f(x) = \left(1 + \frac{1}{x}\right)^x$$

- We were trying to find the value of $f(x)$ as x approaches infinity.

$$\lim_{x \to \infty} f(x) = \lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x = 2.71$$

- This means the limit of $f(x)$, as x approaches infinity, equals 2.71.

Note: This **<u>does not mean</u>** $\boldsymbol{f(x) = 2.71}$ when $\boldsymbol{x = \infty}$**.**

Limits of a function – Definition and Notation

- **<u>Formally, the definition of a limit is:</u>**

$$\lim_{x \to a} f(x) = L$$

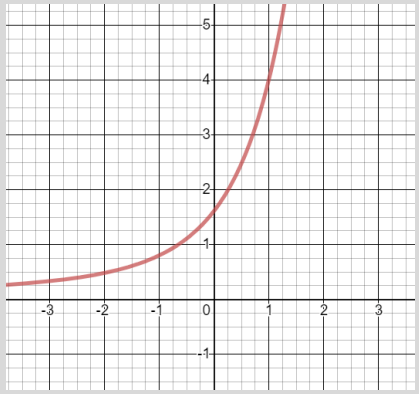
If for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that if $0 < |x - a| < \delta$ then $|f(x) - L| < \varepsilon$.

- **<u>To find the limit of a function:</u>** we substitute into the function with values very close to the limit and see if the results approaches a number. If it does not, the limit of the function is at infinite (positive or negative).

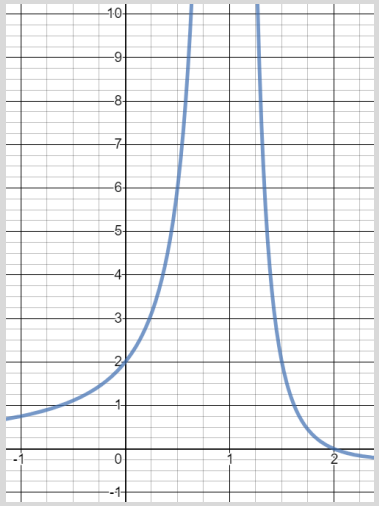
Limits of a function – Definition and Notation Example

Example: Find the limits for the following functions

(a) $\lim_{(x \to 0)} \left(\frac{5^x - 1}{x} \right)$

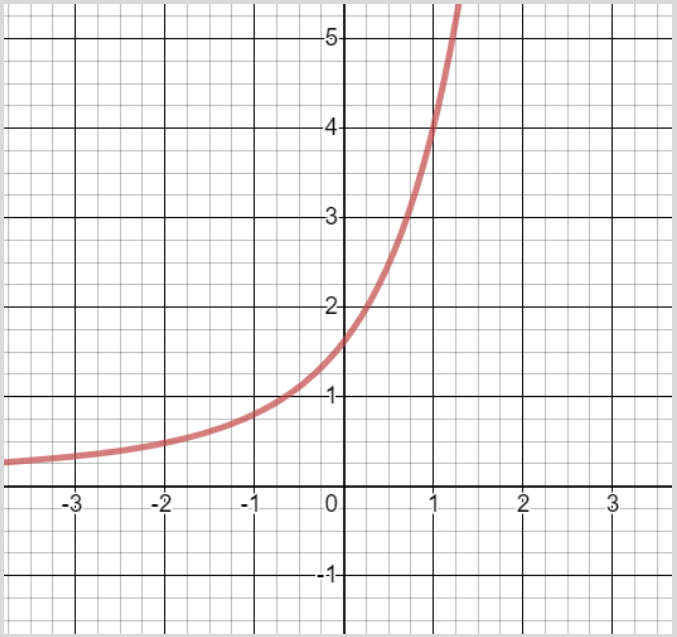


(b) $\lim_{(x \to 1)} \left(\frac{2 - x}{(x-1)^2} \right)$



Limits of a function – Definition and Notation Example

(a) $\lim_{(x \to 0)} \left(\frac{5^x - 1}{x}\right)$



To find the limit value of a function (if it exists), we find the values x that are close to the limit **<u>from both sides</u>** and observe the result. If both the left-hand side and right-hand side limit are equal to each other then we can conclude that the function has a limit value.

(1) From the right-hand side of 0, we shall use 0.1, 0.001, and 0.0001.

$$f(0.1) = \frac{5^{0.1} - 1}{0.1} = 1.746$$

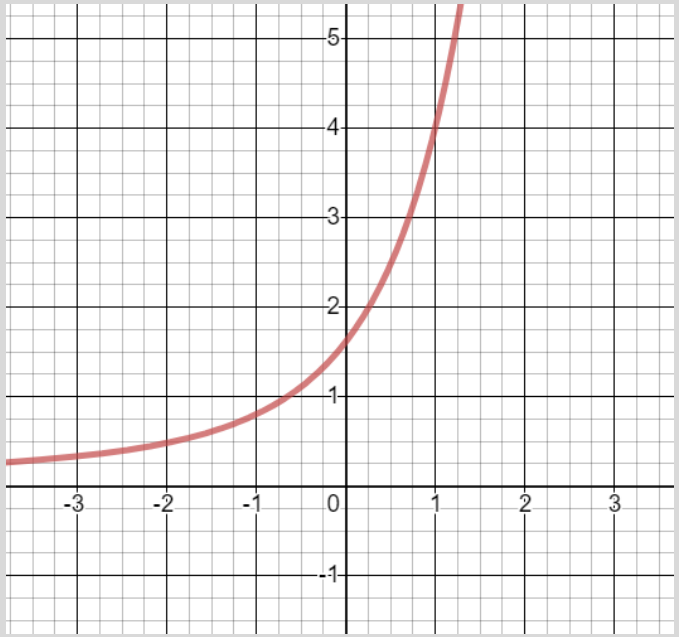
$$f(0.01) = \frac{5^{0.01} - 1}{0.01} = 1.622$$

$$f(0.001) = \frac{5^{0.1} - 1}{0.1} = 1.61$$

It seems the right-hand side limit of f(x) is 1.61.

Limits of a function – Definition and Notation Example

(a) $\lim_{(x \to 0)} \left(\frac{5^x - 1}{x}\right)$



(2) From the left-hand side of 0, we shall use -0.1, -0.001, and -0.0001.

$$f(-0.1) = \frac{5^{-0.1} - 1}{-0.1} = 1.487$$

$$f(-0.01) = \frac{5^{-0.01} - 1}{-0.01} = 1.596$$

$$f(-0.001) = \frac{5^{-0.001} - 1}{-0.001} = 1.61$$

It seems the left-hand side limit of f(x) is 1.61.

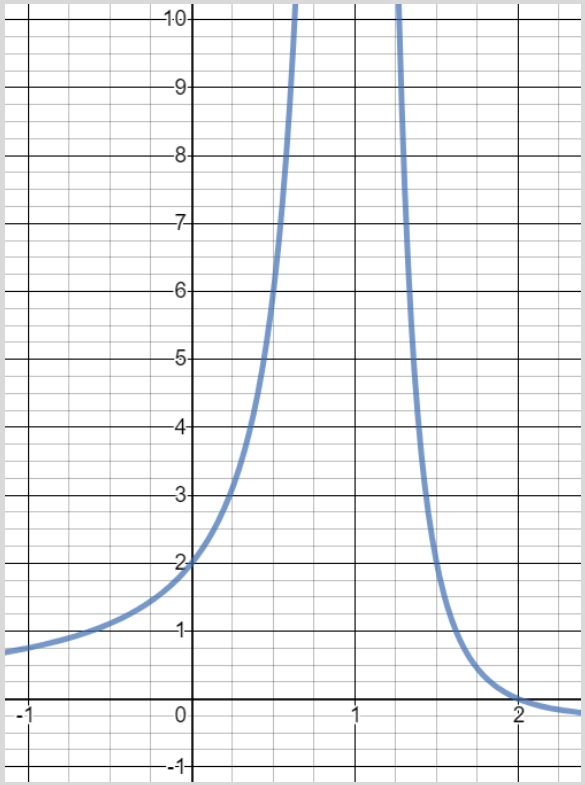
(3) Since the right-hand side limit and the left-hand side limits are the same, we can conclude that:

$$\boldsymbol{\lim_{(x \to 0)} \left(\frac{5^x - 1}{x}\right) = 1.61}$$

Note: This <u>does not mean</u> there is a value $f(0) = 1.61$. The value $f(0)$ does not exist for this function.

Limits of a function – Definition and Notation Example

(b) $\lim_{(x \to 1)} \left(\frac{2-x}{(x-1)^2} \right)$



We shall apply the same procedures as the previous example.

(1) From the right-hand side of 1, we shall use 1.1, 1.01, and 1.001.

$$f(1.1) = \frac{2 - 1.1}{(1.1 - 1)^2} = 90$$

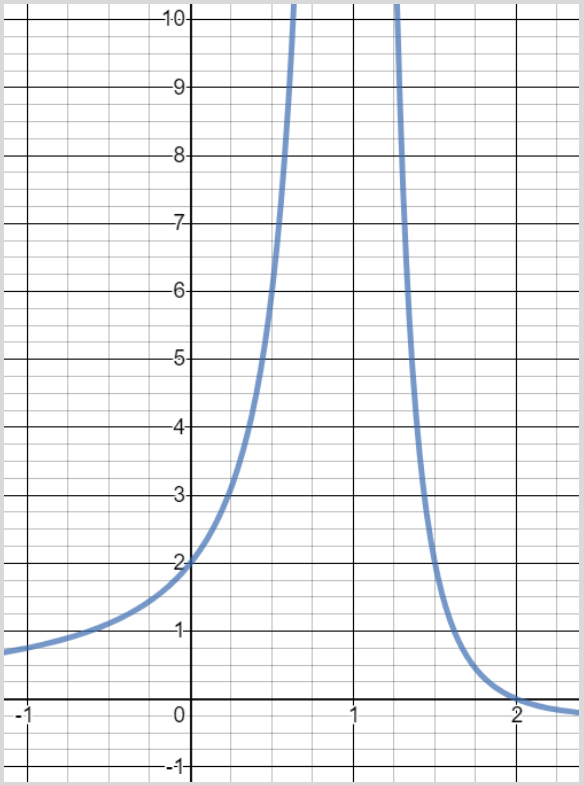
$$f(0.01) = \frac{2 - 1.01}{(1.01 - 1)^2} = 9900$$

$$f(0.001) = \frac{2 - 1.001}{(1.001 - 1)^2} = 999000$$

It seems the right-hand side limit of f(x) is infinity.

Limits of a function – Definition and Notation Example

(b) $\lim_{(x\to 1)} \left(\frac{2-x}{(x-1)^2}\right)$



(2) From the left-hand side of 1, we shall use 0.9, 0.99, and 0.999.

$$f(0.9) = \frac{2-0.9}{(0.9-1)^2} = 110$$

$$f(0.99) = \frac{2-0.99}{(0.99-1)^2} = 10100$$

$$f(0.999) = \frac{2-0.999}{(0.999-1)^2} = 1001000$$

It seems the left-hand side limit of f(x) is also infinity.

(3) Since the right-hand side limit and the left-hand side limits are the same, we can conclude that:

$$\boldsymbol{\lim_{(x\to 0)} \left(\frac{5^x - 1}{x}\right) = \infty}$$

Limits of a function – Calculating Limits

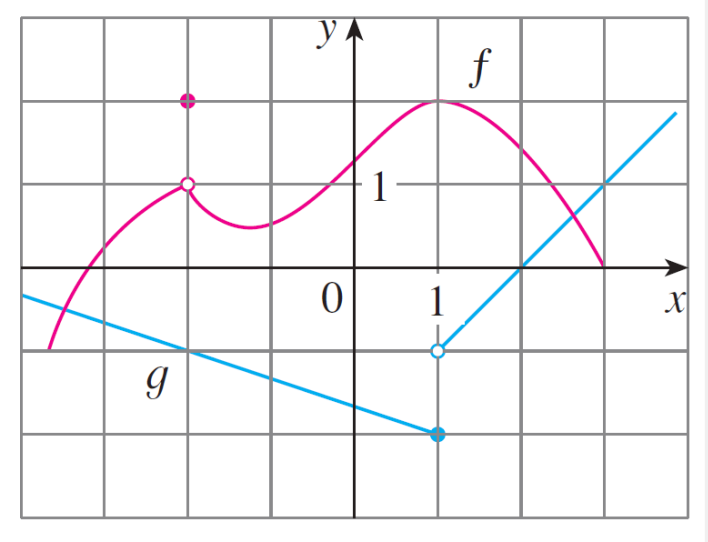
- Given the constant c and the limits $\lim_{x \to a} f(x)$ and $\lim_{x \to a} g(x)$, then the following will be true:
  1. $\lim_{x \to a}[f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$**:** The limit of a sum is the sum of the limits.
  2. $\lim_{x \to a}[f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$**:** The limit of a difference is the difference of the limits.
  3. $\lim_{x \to a}[c * f(x)] = c * \lim_{x \to a} f(x)$**:** the limit of a constant times a function is the constant times the limit of the function.
  4. $\lim_{x \to a}[f(x)g(x)] = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$**:** The limit of a product is the product of the limits.
  5. $\lim_{x \to a}\left(\frac{f(x)}{g(x)}\right) = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$ **if** $\lim_{x \to a} g(x) \neq 0$**:** The limit of a quotient is the quotient of the limits (provided that the limit of the denominator is not 0).

Limits of a function – Calculating Limits Example

<u>Example</u>: Use the Limit Laws and the graphs of f(x) and g(x) below to find the following limits:

(a) $\lim_{x\to -2}[\,f(x) + 5 * g(x)]$

(b) $\lim_{x\to 2}\left(\frac{f(x)}{g(x)}\right)$



(a) $\lim_{x\to -2}[\,f(x) + 5 * g(x)]$

(1) Invoking the Limit Laws, we know that:

$$\lim_{x\to -2}[\,f(x) + 5 * g(x)] = \lim_{x\to -2} f(x) + \lim_{x\to -2} 5 * g(x)$$

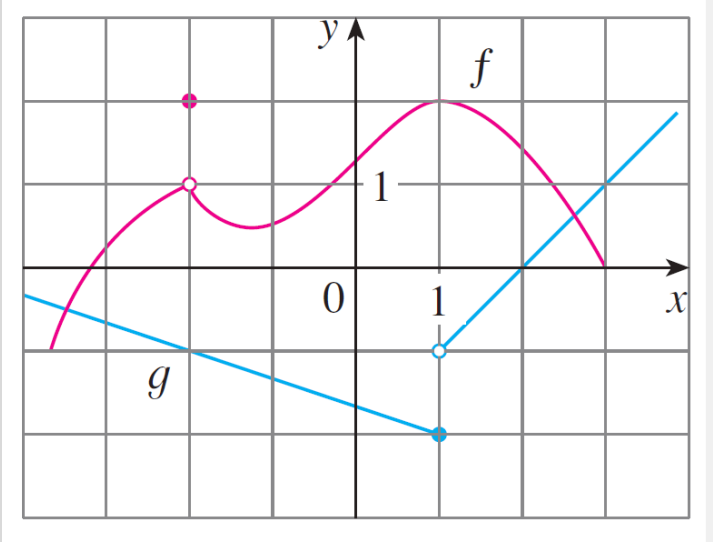
(2) From observing the graph we find that:

$$\lim_{x\to -2} f(x) = 1 \text{ and } \lim_{x\to -2} g(x) = -1$$

(3) Hence:

$$\lim_{x\to -2} f(x) + \lim_{x\to -2} 5 * g(x) = 1 + 5(-1) = \mathbf{-4}$$

Limits of a function – Calculating Limits Example



(b) $\lim_{x\to 2}\left(\frac{f(x)}{g(x)}\right)$

(1) Invoking the Limit Laws, we know that:

$$\lim_{x\to 2}\left(\frac{f(x)}{g(x)}\right) = \frac{\lim_{x\to 2} f(x)}{\lim_{x\to 2} g(x)} \text{ if } \lim_{x\to 2} g(x) \neq 0$$

(2) From observing the graph we find that:

$$\lim_{x\to 2} f(x) \approx 1.4 \text{ and } \lim_{x\to 1} g(x) = 0$$

(3) Since $\lim_{x\to 1} g(x) = 0$, we cannot apply Limit Law #5, thus we must conclude:

$\lim_{x\to 2}\left(\frac{f(x)}{g(x)}\right)$ **does not exist.**

Derivatives and Rates of Change

Splitting the differences

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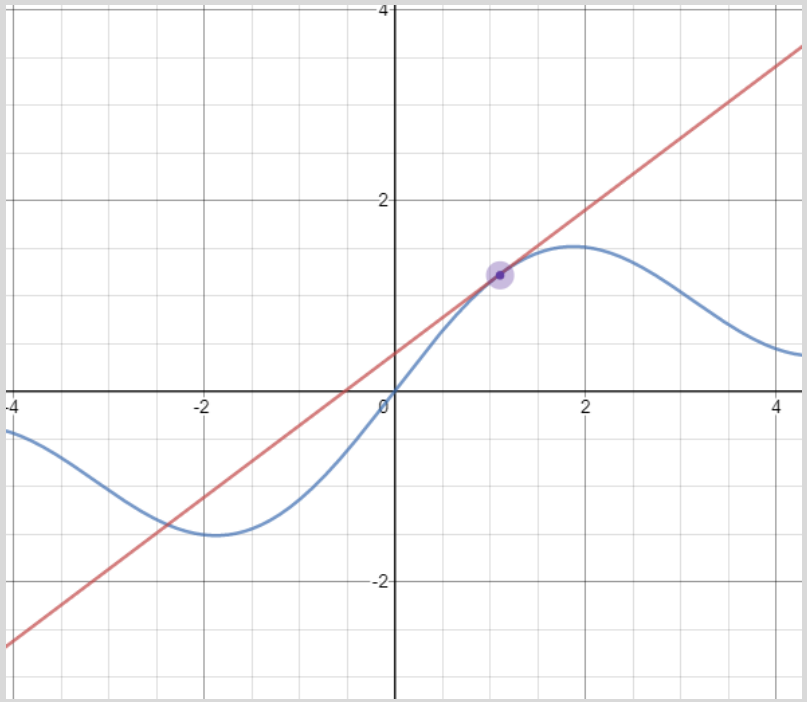
Derivatives – Main topics

- Tangent Lines
- Derivatives – Definition
- Derivative as a Function
  - Notation
  - Differentiability
  - Higher Derivatives

Derivatives – Tangent Lines

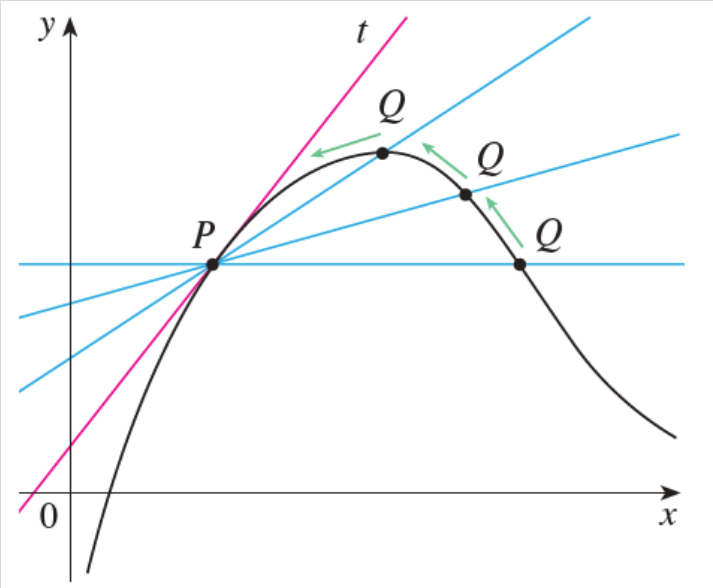
- **<u>Definition:</u>** On any given point of a curve, there exists a line that just **<u>touch</u>** the point without going through the point. This line is called the ***Tangent line***.



Derivatives – Tangent Lines

- **<u>Equation of a line:</u>** The equation of a line passing through any 2 points $A(x_a, y_a)$ and $B(x_b, y_b)$ in 2D Coordinate system can be defined as:

$$\boldsymbol{y = mx + b = \frac{y_b - y_a}{x_b - x_a}(x - x_a) + y_a}$$



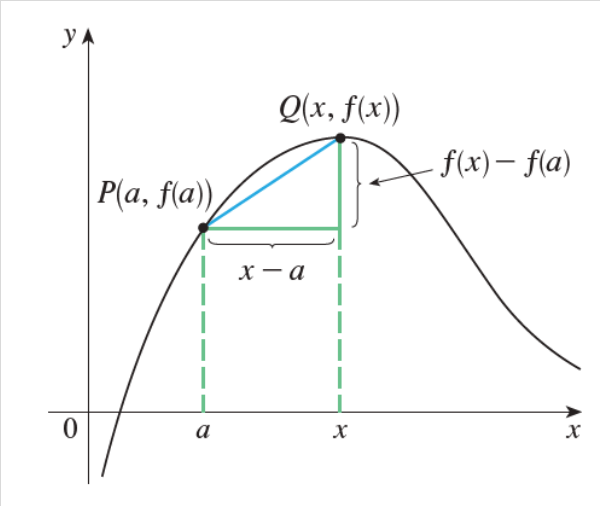
Derivatives – Tangent Lines

- The tangent line can be represented by an equation of the form:

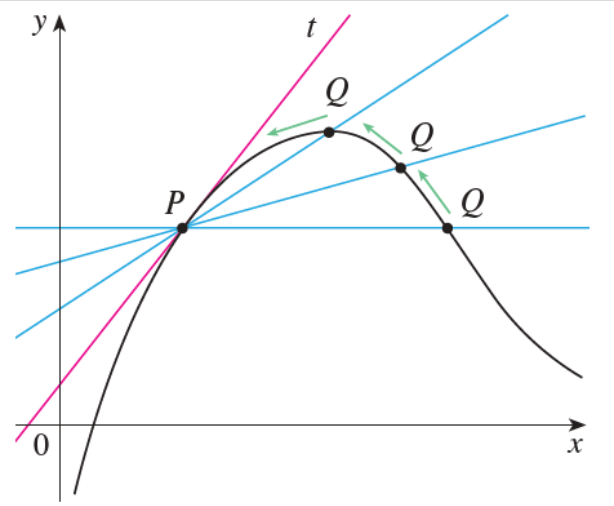
$$\boldsymbol{y = mx + b}$$

Where m = the slope of the line and b = the y-intercept point.

- Given a curve $y = f(x)$, the slope of the tangent line at the point $P(a, f(a))$ can be calculated via the limit:

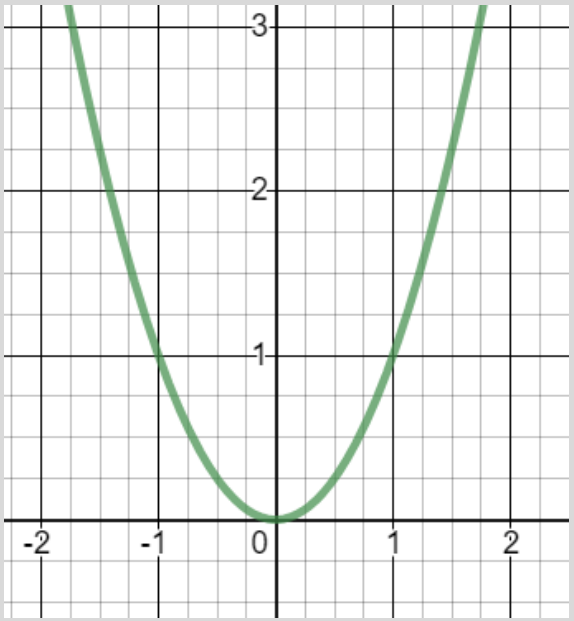


$$\boldsymbol{m = \lim_{x \to a} \left(\frac{f(x) - f(a)}{x - a} \right)}$$



Limits of a function – Calculating Limits Example

<u>Example</u>: Find the equation of the tangent line to the parabola $y = x^2$ at point $P\ (1, 1)$ and plot it.



(1) Invoking the formula to find the slope of the tangent line, we have:

$$m = \lim_{x \to a} \left(\frac{f(x) - f(a)}{x - a} \right)$$

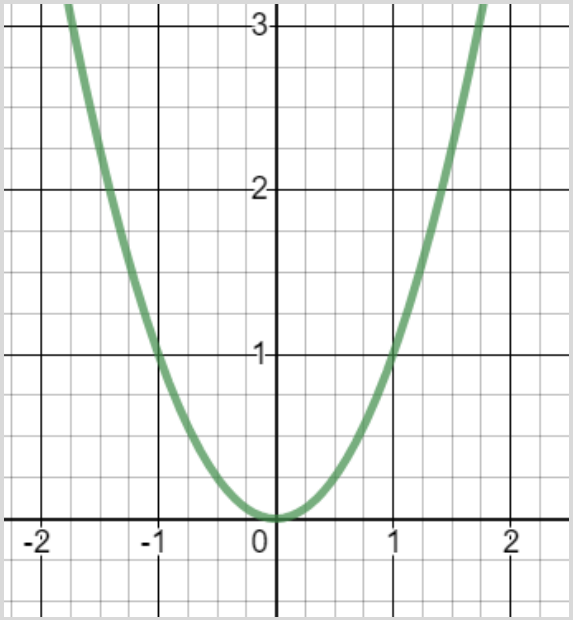
$$\rightarrow m = \lim_{x \to 1} \left(\frac{f(x) - f(1)}{x - 1} \right) = \lim_{x \to 1} \left(\frac{x^2 - 1}{x - 1} \right)$$

$$\rightarrow \lim_{x \to 1} \left(\frac{(x - 1)(x + 1)}{x - 1} \right) = \lim_{x \to 1} x + 1 = 2$$

Thus the slope of the tangent line is $\boldsymbol{m = 2}$ at $P(1,1)$.

Limits of a function – Calculating Limits Example

<u>Example</u>: Find the equation of the tangent line to the parabola $y = x^2$ at point $P\ (1, 1)$.



(2) Invoking the formula of a line in 2D:

$$y = mx + b$$

At point $P(1, 1)$, $x = 1$ and $y\ = 1$, so we have:

$$1 = 2 * 1 + b$$

$$\rightarrow b = 1 - 2 = -1$$

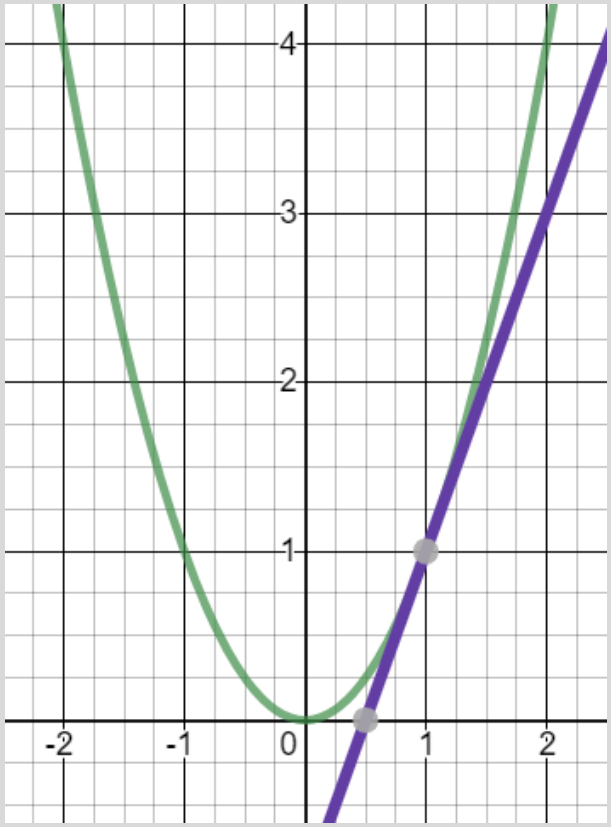
Therefore the y-intercept is point (0, -1).

(3) Write the equation of the tangent line:

$$\boldsymbol{y = 2x - 1}$$

Limits of a function – Calculating Limits Example

<u>Example</u>: Find the equation of the tangent line to the parabola $y = x^2$ at point $P\ (1, 1)$ and plot it.



(4) Plot the line on 2D Cartesian Coordinates:

$$y = 2x - 1$$

Derivatives – Definition

- There is another way to write the formula to calculate the slope of the tangent line in 2D Cartesian Coordinate system at any point $P\ (a, f(a))$:

$$\boldsymbol{m = \lim_{h \to 0} \left(\frac{f(a+h) - f(a)}{h} \right)}$$

- The formula above is also called the Derivative of a function $f(x)$ at a number a, written as $f'(a)$ (pronounced $f\ prime\ of\ a$)

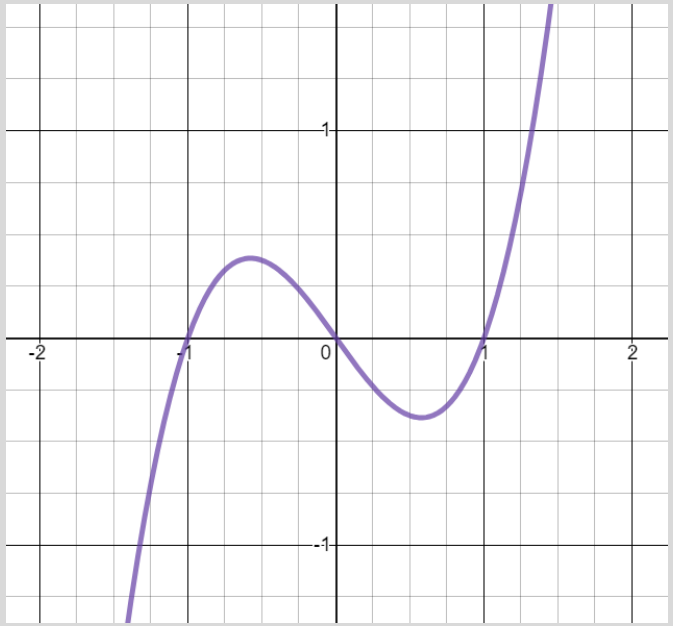
$$\boxed{\boldsymbol{f'(a) = \lim_{h \to 0} \left(\frac{f(a+h) - f(a)}{h} \right) = \lim_{x \to a} \left(\frac{f(x) - f(a)}{x - a} \right)}}$$

where $\boldsymbol{x = a + h}$.

- **The Derivative $\boldsymbol{f'(a)}$ is also the instantaneous rate of change of $\boldsymbol{y = f(x)}$ with respect to $\boldsymbol{x}$ when $\boldsymbol{x = a}$.**

Derivatives – Example

<u>Example</u>: Find the derivative $f'(a)$ of $f(x) = x^3 - x$ where a is a real number.



Invoking the formula to find the derivative of a function, we have:

$$f'(a) = \lim_{h \to 0} \left(\frac{f(a+h) - f(a)}{h} \right)$$

$$\rightarrow f'(a) = \lim_{h \to 0} \left(\frac{(a+h)^3 - (a+h) - [a^3 - a]}{h} \right)$$

$$\rightarrow \lim_{h \to 0} \left(\frac{a^3 + 3a^2h + 3ah^2 + h^3 - a - h - a^3 + a}{h} \right)$$

$$\rightarrow \lim_{h \to 0} \left(\frac{3a^2h + 3ah^2 + h^3 - h}{h} \right)$$

$$\rightarrow \lim_{h \to 0} (3a^2 + 3ah + h^2 - 1) = \mathbf{3a^2 - 1}$$

We can take the general case of $x = a$ and write that: $f(x) = 3x^2 - 1$

Derivatives – Notations

- Some common notations of a derivative is written below:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = Df(x) = D_x f(x)$$

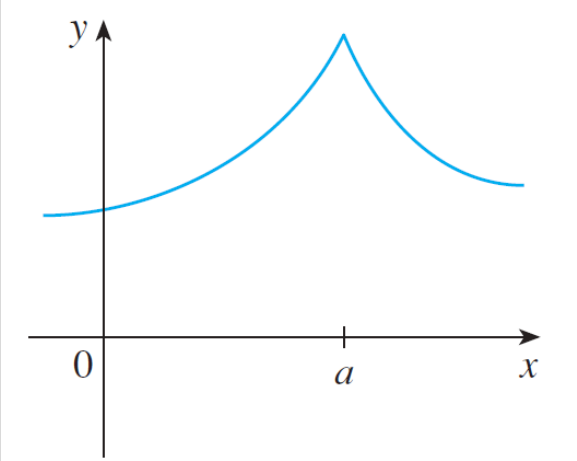
- **The notation $\frac{dy}{dx}$ is not a ratio**, it is to be read as "differentiation of y with respect to x". The definition of $\frac{dy}{dx}$ can be written as:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

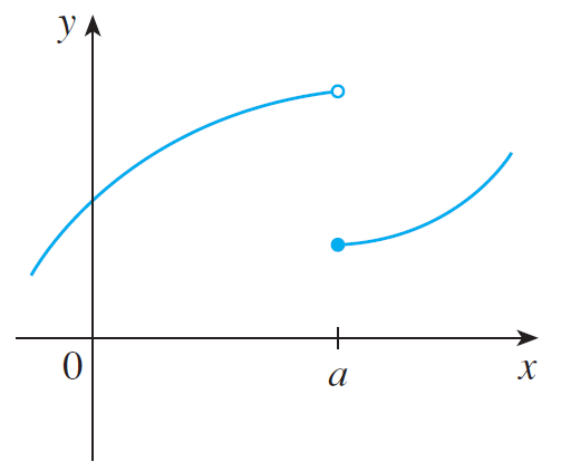
Note: $\frac{dy}{dx}$ is also called the Leibniz notation and it is very popular in engineering.

Derivatives – Differentiability

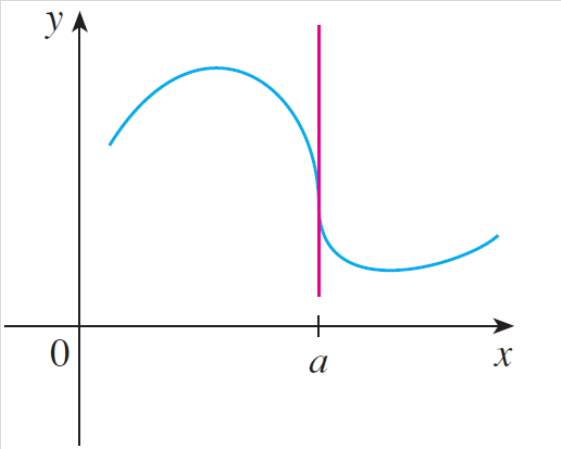
- Not all $f(x)$ functions are differentiable at all values of x.
- There are 3 cases where a function is not differentiable:



(a) A corner



(b) A discontinuity



(c) A vertical tangent

Derivatives – Higher Derivative

- f' may have its own derivative $(f')' = f''$, which is called the second derivative of f.
- The Leibniz notation of the second derivative is:

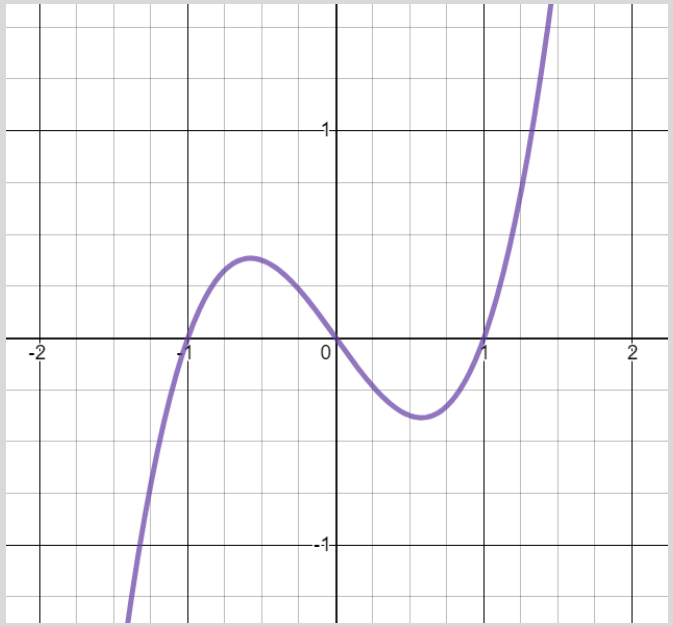
$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}$$

- Hence, we also have the nth derivate of f which is written as:

$$f^n(x) = \frac{d^ny}{dx^n}$$

Derivatives – Example

<u>Example</u>: Find the 2nd derivative $f''(x)$ of $f(x) = x^3 - x$. Plot $f(x), f'(x), f''(x)$ on the same graph



(1) Recall from the previous example:

$$f'(x) = 3x^2 - 1$$

(2) Invoke the formula to find the derivative of a function:

$$f''(x) = \lim_{h \to 0} \left(\frac{f'(x+h) - f'(x)}{h} \right)$$

$$= \lim_{h \to 0} \left(\frac{3(x+h)^2 - 1 - [3x^2 - 1]}{h} \right)$$

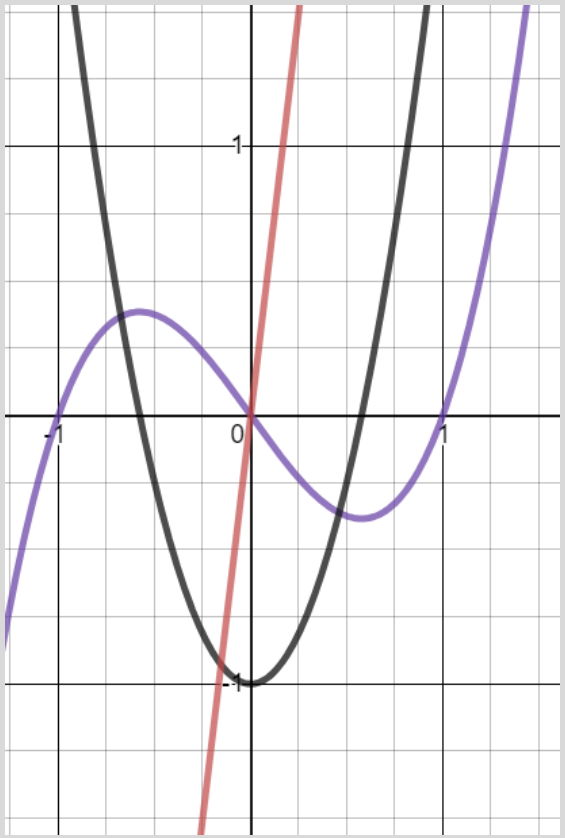
$$= \lim_{h \to 0} \left(\frac{3(x^2 + 2xh + h^2) - 1 - 3x^2 + 1]}{h} \right)$$

$$= \lim_{h \to 0} \left(\frac{3x^2 + 6xh + 3h^2 - 3x^2}{h} \right) = \lim_{h \to 0} (3h + 6x)$$

$$\underline{= \boldsymbol{6x}}$$

Derivatives – Example

<u>Example</u>: Find the 2nd derivative $f''(x)$ of $f(x) = x^3 - x$. Plot $f(x), f'(x), f''(x)$ on the same graph



(3) Plot the functions

$$\boldsymbol{f(x) = x^3 - x}$$

$$\boldsymbol{f'(x) = 3x^2 - 1}$$

$$\boldsymbol{f''(x) = 6x}$$

Derivatives – Section Highlights

- The derivative of a function at a point on a curve will yield **the slope of the tangent line** to the curve through that point.
- There are 3 cases where taking **a derivative is not possible**.
  - A corner.
  - A discontinuity.
  - A vertical tangent.

Differentiation Formulas

The shortcuts

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Differentiation Formulas– Main topics

1. **Derivative of a Constant.**
2. **Power Rule**
3. **Constant Multiple Rule**
4. **Sum rule**
5. **Difference Rule**
6. **Product Rule**
7. **Quotient Rule**
8. **Chain Rule.**
    - **Implicit differentiation**
    - **Parametric differentiation**

Differentiation Formulas– Introduction

- Recall the formula of Derivatives:

$$f'(x) = \lim_{h \to 0} \left(\frac{f(x+h) - f(x)}{h} \right)$$

- In engineering problems, we mostly <u>will not use the formula above</u> and rely on **<u>8 formal rules</u>** of Derivatives to solve problems.

Differentiation Formulas– Derivative of a constant Function

- **<u>Rule 1: Derivative of a constant function:</u>**
  - Given a constant c, then the derivative of c is:

$$\boldsymbol{f'(x) = \lim_{h\to 0}\left(\frac{f(x+h)-f(x)}{h}\right) = \lim_{h\to 0}\left(\frac{c-c}{h}\right) = 0}$$

Thus we can conclude that:

$$\boldsymbol{\frac{d}{dx}(c) = 0}$$

<u>Example:</u> If $f(x) = 3$, then $f'(x) = \frac{d}{dx}(3) = 0$.

Differentiation Formulas– Power Rule

- **<u>Rule 2: Power Rule:</u>**
  - Recall the previous example: $f(x) = x^3 - x$ and $f'(x) = 3x^2 - 1$ and $f''(x) = 6x$. Notice that every time we take a derivative, the power of x decreases by 1.
  - Thus we have the general case:

$$\boldsymbol{\frac{d}{dx}(x^n) = n * x^{n-1}}$$

<u>Example:</u> Find the derivative of $f(x) = x^3$.

Invoking the Power Rule we find that:

$$f'(x) = \frac{d}{dx}(4x^3) = 3 * x^{3-1} = 3x^2$$

Note: In the case of $f(x) = x$, then $f'(x) = 1 * x^{1-1} = 1$.

Differentiation Formulas– Constant Multiple Rule

- **<u>Rule 3: Constant multiple rule:</u>**
  - Recall Law #3 of the Limit Laws: $\lim_{x \to a}[c * f(x)] = c * \lim_{x \to a} f(x)$.
  - Applying this Law to the case where $g(x) = cf(x)$, thus

$$g'(x) = \lim_{h \to 0}\left(\frac{cf(x+h) - cf(x)}{h}\right) = c \lim_{h \to 0}\left(\frac{f(x+h) - f(x)}{h}\right)$$

  - Hence, we can conclude the general case:

$$\frac{d}{dx}[cf(x)] = c * \frac{d}{dx} f(x)$$

<u>Example:</u> find the derivative of $f(x) = 4x^3$

The derivative is $f'(x) = 4 * (3 * x^{3-1}) = 12x^2$

Differentiation Formulas– The Sum Rule

- **<u>Rule 4: The Sum Rule:</u>**
  - Recall Law #1 of the Limit Laws: $\boldsymbol{\lim_{x\to a}[f(x)+g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)}$.
  - Applying this Law to the case where $F(x) = f(x) + g(x)$, thus

$$F'(x) = \lim_{h\to 0}\left(\frac{[f(x+h)+g(x+h)] - [f(x)+g(x)]}{h}\right)$$

$$\to F'(x) = \lim_{h\to 0}\left(\frac{f(x+h)-f(x)}{h}\right) + \lim_{h\to 0}\left(\frac{g(x+h)+g(x)}{h}\right)$$

  - Hence, we can conclude the general case:

$$\boldsymbol{\frac{d}{dx}[f(x)+g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)}$$

<u>Example:</u> find the derivative of $f(x) = 4x^3 + 2x^2$

The derivative is $f'(x) = 4 * (3 * x^{3-1}) + 2 * (2 * x^{2-1}) = 12x^2 + 4x$

Differentiation Formulas– The Difference Rule

- **<u>Rule 5: The Difference Rule:</u>**
  - Recall Law #2 of the Limit Laws: $\boldsymbol{\lim_{x\to a}[f(x)-g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)}$.
  - Applying this Law to the case where $F(x) = f(x) - g(x)$, thus

$$F'(x) = \lim_{h\to 0}\left(\frac{[f(x+h)-g(x+h)]-[f(x)-g(x)]}{h}\right)$$

$$\to F'(x) = \lim_{h\to 0}\left(\frac{f(x+h)-f(x)}{h}\right) - \lim_{h\to 0}\left(\frac{g(x+h)-g(x)}{h}\right)$$

  - Hence, we can conclude the general case:

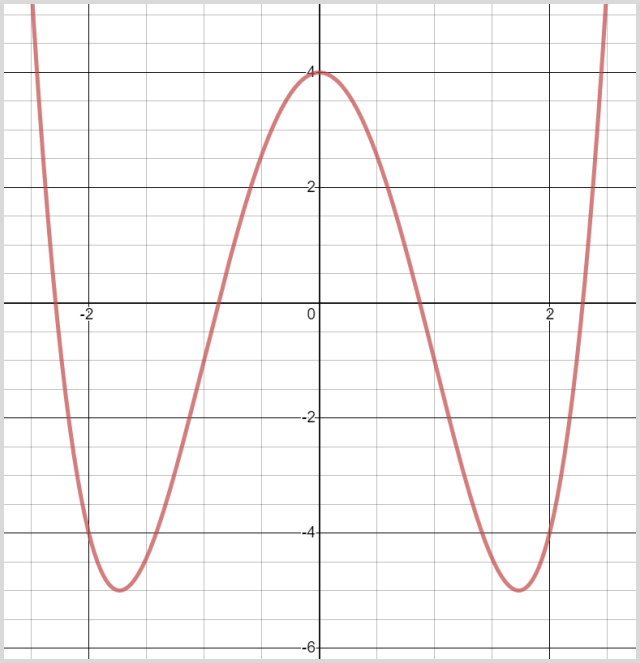
$$\boldsymbol{\frac{d}{dx}[f(x)-g(x)] = \frac{d}{dx}f(x) - \frac{d}{dx}g(x)}$$

<u>Example:</u> find the derivative of $f(x) = 4x^3 - 2x^2$

The derivative is $f'(x) = 4 * (3 * x^{3-1}) - 2 * (2 * x^{2-1}) = 12x^2 - 4x$

Differentiation Formulas– Example 1

- **<u>Example 1:</u>** find the coordinates of the points on the curve $\mathrm{f(x)} = y = x^4 - 6x^2 + 4$ where the tangent line is <u>horizontal</u>.



(1) Recall that a horizontal line in 2D Coordinate system has the slope of 0.

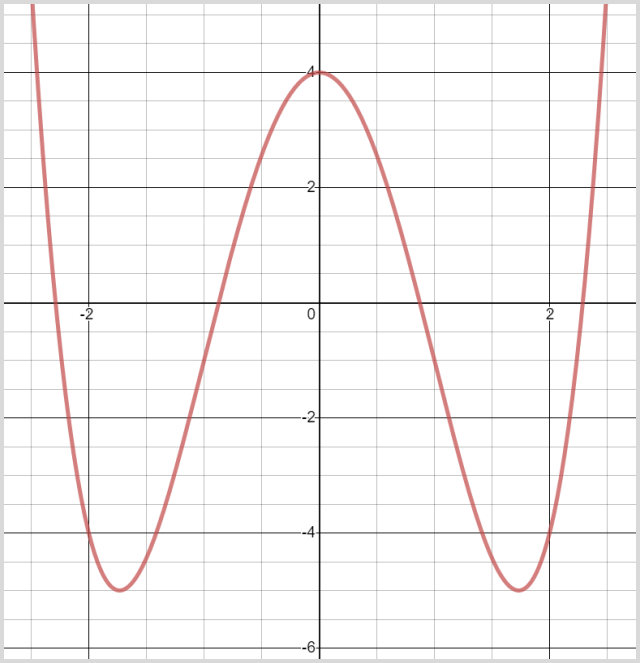
(2) Recall that the slope of a tangent line to any curve is provided by the derivative of that curve.

(3) First, we find the derivative $f'(x)$:

$$f'(x) = \frac{d}{dx}(x^4 - 6x^2 + 4)$$
$$= \frac{d}{dx}(x^4) - \frac{d}{dx}(6x^2) + \frac{d}{dx}(4)$$
$$= 4(x^{4-1}) - 6 * (2x^{2-1} + 0$$
$$= \boldsymbol{4x^3 - 12x}$$

Differentiation Formulas– Example 1

- **<u>Example 1:</u>** find the coordinates of the points on the curve $\mathrm{f(x)} = y = x^4 - 6x^2 + 4$ where the tangent line is <u>horizontal</u>.



(4) With $f'(x) = 4x^3 - 12x$, we can find the slope of the tangent point at any x-coordinate on the curve. However, we are interested only in points where $f'(x) = 0$ where the tangent line is horizontal. Thu we solve the equation:

$$f'(x) = 0 = 4x^3 - 12 = 4x(x^2 - 3)$$

The equation above has 3 roots:

$$x_1 = 0, x_2 = \sqrt{3}, x_3 = -\sqrt{3}$$

(5) Finally we use the function $f(x)$ to find their y-coordinates:

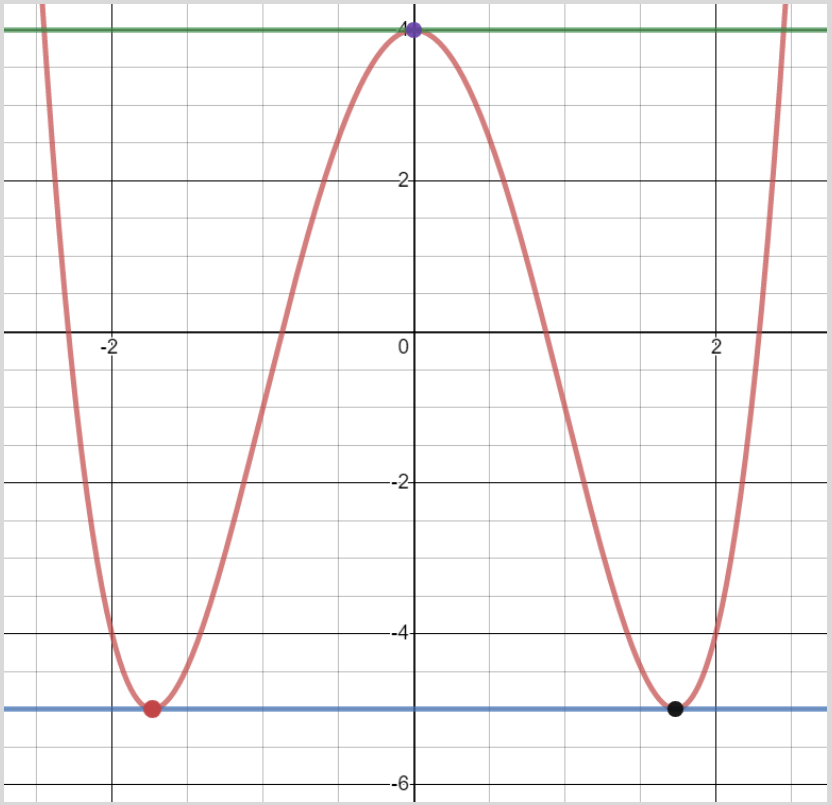
$y_1 = f(0) = 0^4 - 6 * 0^2 + 4 = 4$

$y_2 = f(\sqrt{3}) = \sqrt{3}^4 - 6 * \sqrt{3}^2 + 4 = -5$

$y_3 = f(-\sqrt{3}) = (-\sqrt{3})^4 - 6 * (-\sqrt{3})^2 + 4 = 5$

Differentiation Formulas– Example 1

- **<u>Example 1:</u>** find the coordinates of the points on the curve $\mathrm{f}(\mathrm{x}) = y = x^4 - 6x^2 + 4$ where the tangent line is <u>horizontal</u>.



(6) We conclude that there are 3 points where the tangent lines are horizontal for $f(x)$ which are:

$\boldsymbol{P_1(0,4); P_2(\sqrt{3},-5);}$
$\boldsymbol{P_3(-\sqrt{3},-5)}$

Differentiation Formulas– Product Rule

- **Rule 6: The Product Rule:**
  - Given $F(x) = f(x) * g(x)$, thus

$$\boldsymbol{F'(x) = \lim_{h\to 0}\left(\frac{[f(x+h) * g(x+h)] - [f(x) * g(x)]}{h}\right)}$$

Here we add a term $[f(x+h) * g(x) - f(x+h) * g(x)]$ to the numerator:

$$F'(x)$$

$$= \lim_{h\to 0}\left(\frac{[f(x+h) * g(x+h)] + f(x+h) * g(x) - f(x+h) * g(x) - [f(x) * g(x)]}{h}\right)$$

$$= \lim_{h\to 0}\left(g(x)\frac{f(x+h) - f(x)}{h}\right) + \lim_{h\to 0}\left((f(x+h)\frac{g(x+h) - g(x)}{h}\right)$$

  - Hence, we can conclude the general case:

$$\boldsymbol{\frac{d}{dx}[f(x) * g(x)] = g(x)\frac{d}{dx}f(x) + f(x)\frac{d}{dx}g(x)}$$

Differentiation Formulas– Product Rule Example

- **<u>The Product Rule:</u>**

$$\boldsymbol{\frac{d}{dx}[f(x) * g(x)] = g(x)\frac{d}{dx}f(x) + f(x)\frac{d}{dx}g(x)}$$

<u>Example:</u> Given $F(x) = (6x^3)(7x^4)$, find $F'(x)$

Invoking Product Rule, we know that:

$$F'(x) = (7x^4)\frac{d}{dx}(6x^3) + (6x^3)\frac{d}{dx}(7x^4)$$

$$= (7x^4)(6 * 3x^2) + (6x^3)(7 * 4x^3)$$

$$= (168x^6 + 126x^6) = \boldsymbol{294x^6}$$

There is another way to check our answer. We recognize that:

$$F(x) = (6x^3)(7x^4) = 56x^7$$

$$F'(x) = 56 * 7x^6 = \boldsymbol{294x^6}$$

Differentiation Formulas– Quotient Rule

- **<u>Rule 7: The Quotient Rule:</u>**
  - Given $F(x) = \frac{f(x)}{g(x)}$, thus

$$\boldsymbol{F'^{(x)} = \lim_{h\to 0}\left(\frac{\left[\frac{f(x+h)}{g(x+h)}\right] - \left[\frac{f(x)}{g(x)}\right]}{h}\right)}$$

$$= \lim_{h\to 0}\left(\frac{[f(x+h) * g(x)] - [f(x) * g(x+h)]}{h * g(x+h)g(x)}\right)$$

Here we add a term $[f(x) * g(x) - f(x) * g(x)]$ to the numerator:

$$F'(x) = \lim_{h\to 0}\left(\frac{[f(x+h) * g(x)] + f(x) * g(x) - f(x) * g(x) - [f(x) * g(x+h)]}{h * g(x+h)g(x)}\right)$$

$$= \lim_{h\to 0}\frac{\left(g(x)\frac{f(x+h) - f(x)}{h}\right) - \left(f(x)\left(\frac{g(x+h) - g(x)}{h}\right)\right)}{g(x+h)g(x)} = \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}$$

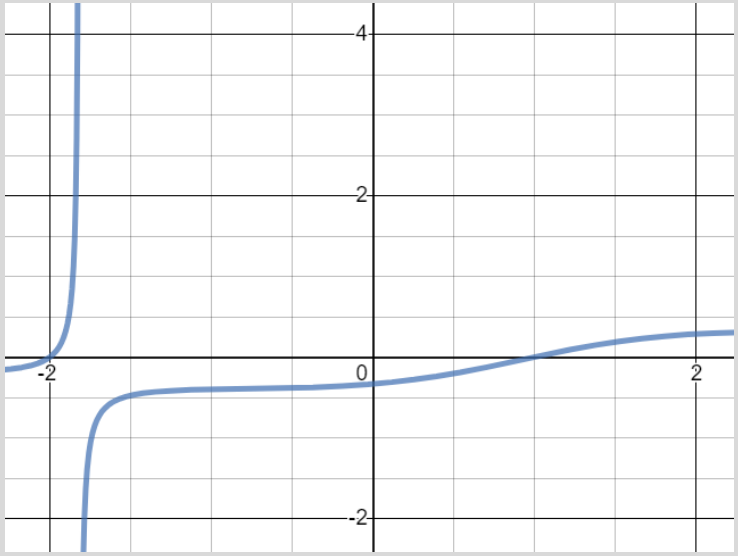
Differentiation Formulas– Quotient Rule

- **<u>Rule 7: The Product Rule:</u>**
  - Hence, we can conclude the general case:

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}[f(x)] - f(x)\frac{d}{dx}[g(x)]}{[g(x)]^2}$$

Differentiation Formulas– Quotient Rule Example

Example: Given $F(x) = \frac{x^2+x-2}{x^3+6}$, find $F'(x)$



Invoking the Quotient Rule

$$F'(x)$$
$$= \frac{(x^3+6)\frac{d}{dx}(x^2+x-2) - (x^2+x-2)\frac{d}{dx}(x^3+6)}{(x^3+6)^2}$$
$$= \frac{(x^3+6)(2x+1) - (x^2+x-2)(3x^2)}{(x^3+6)^2}$$
$$= \frac{(2x^4+x^3+12x+6) - (3x^4+3x^3-6x^2)}{(x^3+6)^2}$$
$$= \frac{-x^4-2x^3+6x^2+12x+6}{(x^3+6)^2}$$

Note: there is a discontinuity on this curve at $x = \sqrt[3]{-6} = -1.817$. At that location, the function is indifferentiable.

Differentiation Formulas– Chain Rule

Rule 8: The Chain Rule:

- The **Chain Rule is one of the most important rules** and where people often make mistakes.
- The rule provides a way for us to find derivative of a function which has another function within it. **Example**: $F(x) = f(g(x)) = \sqrt{x^2 + 1}$

Differentiation Formulas– Chain Rule

Rule 8: The Chain Rule:

- Recall one of the definitions of derivative:

$$\boldsymbol{\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}}$$

Where Δy and Δx are real numbers with values.

In the case of $f\big(g(x)\big)$, let's define a quantity $\boldsymbol{u = g(x)}$ and $\Delta u = g(x + \Delta x) - g(x)$, hence we have:

$$y = f(u)$$

$$\boldsymbol{\Delta y} = f(u + \Delta u) - f(u)$$

$$\rightarrow \frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta u} . \frac{\Delta u}{\Delta x} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta u} . \lim_{\Delta x \to 0} \frac{\Delta u}{\Delta x}$$

$$\rightarrow \frac{dy}{dx} = \frac{dy}{du} . \frac{du}{dx} = \boldsymbol{f'\big(g(x)\big) . g'(x)}$$

Differentiation Formulas– Chain Rule

Rule 8: The Chain Rule:

- Hence, we can conclude the general case for any function $F(x) = f(g(x))$:

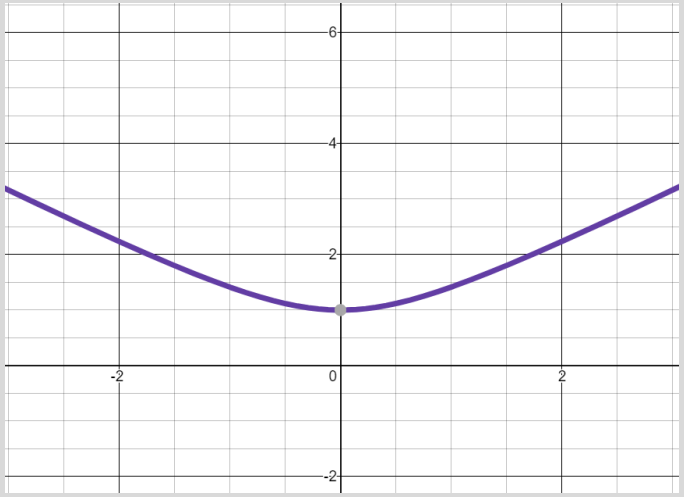
$$\boldsymbol{F'(x) = f'\big(g(x)\big).\, g'(x)}$$

- Combining the expression above with Power rule, we can conclude that, for any $u = g(x)$ and $y = f(u)$:

$$\boldsymbol{f'(u(x)) = \frac{d}{dx}(u^n) = n * u^{n-1}\frac{du}{dx}}$$

Differentiation Formulas– Chain Rule Example

Example: Given $F(x) = \sqrt{x^2 + 1}$, find $F'(x)$



(1) We can rewrite the function as follows:

$$F(x) = (x^2 + 1)^{\frac{1}{2}}$$

(2) Thus we can identify the 2 functions:

$$u = x^2 + 1 \text{ and } f(u) = u^{\frac{1}{2}}$$

(3) We can then invoke the Chain Rule to find the derivative of the function $F(x)$:

$$\begin{aligned} F'(x) &= \frac{d}{du}(u)^{\frac{1}{2}} * \frac{d}{dx}(x^2 + 1) \\ &= \frac{1}{2}(u)^{\frac{1}{2}-1} * (2x) \\ &= \frac{1}{2}(x^2 + 1)^{-\frac{1}{2}} * 2x \\ &= \boldsymbol{\frac{x}{\sqrt{x^2 + 1}}} \end{aligned}$$

Differentiation Formulas– Chain Rule & Implicit Differentiation

Implicit Differentiation:

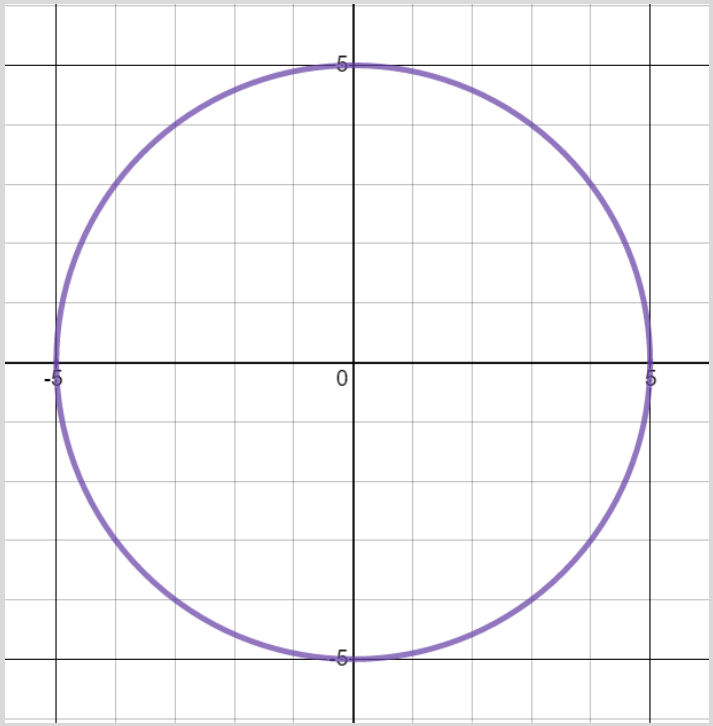
- In engineering and physics, there are some functions that are defined implicitly by a relation between x and y, instead of y in terms of x, such as:

$$x^2 + y^2 = 25 \ (equation\ of\ a\ circle)$$

Differentiation Formulas– Chain Rule & Implicit Differentiation

Implicit Differentiation Example:

Example: Find the derivative $\frac{dy}{dx}$ knowing that: $x^2 + y^2 = 25$. What is the slope of the tangent line at point $P(3,4)$



(1) We take the derivative of the entire equation with respect to x:

$$\frac{d}{dx}(x^2 + y^2 = 25) \leftrightarrow \frac{dx^2}{dx} + \frac{dy^2}{dx} = \frac{d}{dx}25$$

$$\rightarrow 2x + \frac{d}{dy}y^2 . \frac{dy}{dx} = 0$$

$$\rightarrow 2x + 2y\frac{dy}{dx} = 0$$

$$\rightarrow \boldsymbol{\frac{dy}{dx} = -\frac{x}{y}}$$

(2) At point $P(3,4)$, $x = 3$ and $y = 4$, so the slope is:

$$\boldsymbol{\frac{dy}{dx} = -\frac{3}{4}}$$

Differentiation Formulas–
Chain Rule & Parametric Differentiation

Parametric Differentiation:

- Recall from Vector's Lecture, we discussed parametric equations for a line which has the form:

$$x(t) = -1 + 2 * t$$
$$y(t) = 7 - 3 * t$$
$$z(t) = -10 + 4 * t$$

- The Chain Rule allows us to perform Parametric Differentiation on these functions to obtain $\frac{dy}{dx}$, as follows:

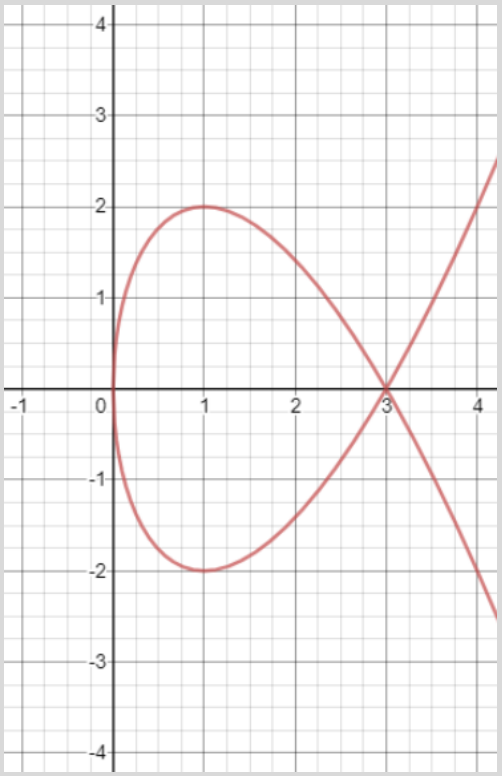
$$\boldsymbol{\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\frac{dx}{dt}}}$$

Note: Only valid if $\frac{dx}{dt} \neq 0$.

Differentiation Formulas–
Chain Rule & Parametric Differentiation

Parametric Differentiation Example:

Example: Find the derivative $\frac{dy}{dx}$ of the parametric curve $y = t^3 - 3t$ and $x = t^2$. Find the slope(s) of the tangent line to the curve at point (3, 0).



(1) We invoke the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\frac{dx}{dt}}$$

(2) We recognize the derivatives of x and y with respect to the parameter t as:

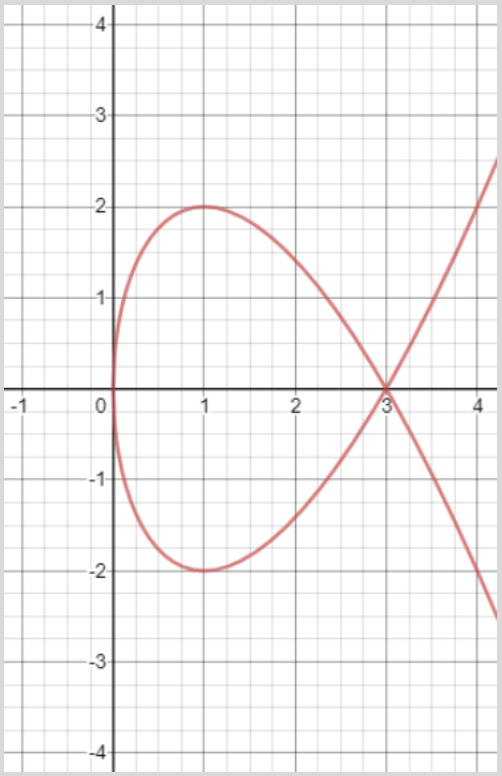
$$\frac{dy}{dt} = \frac{d}{dt}(t^3 - 3t) = 3t^2 - 3$$

$$\frac{dx}{dt} = \frac{d}{dt}(t^2) = 2t$$

Differentiation Formulas–
Chain Rule & Parametric Differentiation

Parametric Differentiation Example:

Example: Find the derivative $\frac{dy}{dx}$ of the parametric curve $y = t^3 - 3t$ and $x = t^2$. Find the slope(s) of the tangent line to the curve at point (3, 0).



(3) Substituting the functions from (2) to the formula in (1), we get :

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\frac{dx}{dt}} = \frac{3t^2 - 3}{2t} = \boldsymbol{\frac{3}{2}\left(t - \frac{1}{t}\right)}$$

(4) Next we find the parameter value t at point (3, 0). Recall:

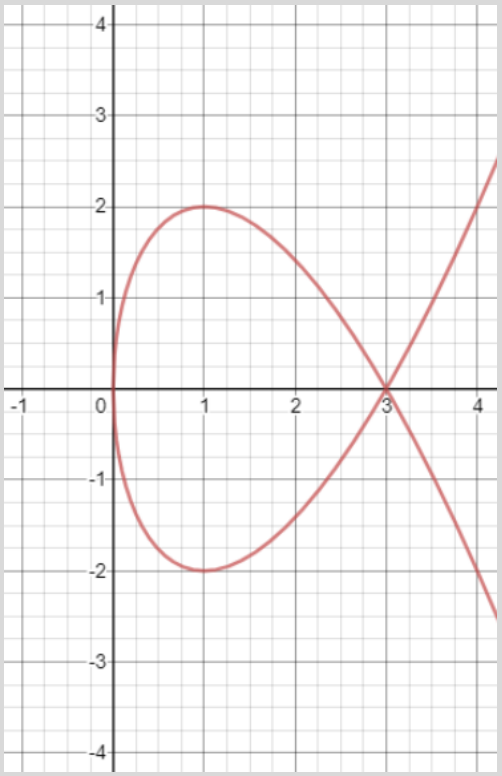
$$y = t^3 - 3t \ \rightarrow y = t(t^2 - 3)$$
$$\rightarrow \mathbf{0} = \boldsymbol{t}(\boldsymbol{t}^{\mathbf{2}} - \mathbf{3})$$

The equation above has 3 roots which are $\boldsymbol{t} = \mathbf{0}, \boldsymbol{t} = \sqrt{\mathbf{3}}, \boldsymbol{t} = -\sqrt{\mathbf{3}}$. The case $t = 0$ does not count so the curve has **2 tangent lines** at point $(3, 0)$.

Differentiation Formulas–
Chain Rule & Parametric Differentiation

Parametric Differentiation Example:

Example: Find the derivative $\frac{dy}{dx}$ of the parametric curve $y = t^3 - 3t$ and $x = t^2$. Find the slope(s) of the tangent line to the curve at point (3, 0).



(5) Use the parametric values from (4) and the derivative from (3) to find the slopes of the tangent lines:

$$t = \sqrt{3} \rightarrow \frac{dy}{dx} = \frac{3}{2}\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = \boldsymbol{\sqrt{3}}$$

$$t = -\sqrt{3} \rightarrow \frac{dy}{dx} = \frac{3}{2}\left(-\sqrt{3} + \frac{1}{\sqrt{3}}\right) = \boldsymbol{-\sqrt{3}}$$

Differentiation Formulas – Exercises

Find the derivatives $\frac{dy}{dx}$ for the following functions:

1. $f(x) = \sqrt{5x} + \frac{\sqrt{7}}{x}$

2. $f(x) = \frac{\sqrt{x}+x}{x^2}$

3. $f(x) = (1 + x + x^2)^{99}$

4. $x^4 + x^2y^2 + y^2 = 5$

Derivatives of Special Functions

Identities you should know

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Derivatives of Special functions– Main topics

1. Derivatives of Trigonometric Functions
2. Derivatives of Inverse Trigonometric functions
3. Derivatives of Exponential Functions
4. Derivatives of Logarithm Functions
5. Derivatives of Hyperbolic Functions

Derivatives of Trigonometric functions

- For $y = \sin(x)$:

$$\frac{d}{dx}\sin(x) = sin(x).0 + \cos(x).1 = \cos(\mathrm{x})$$

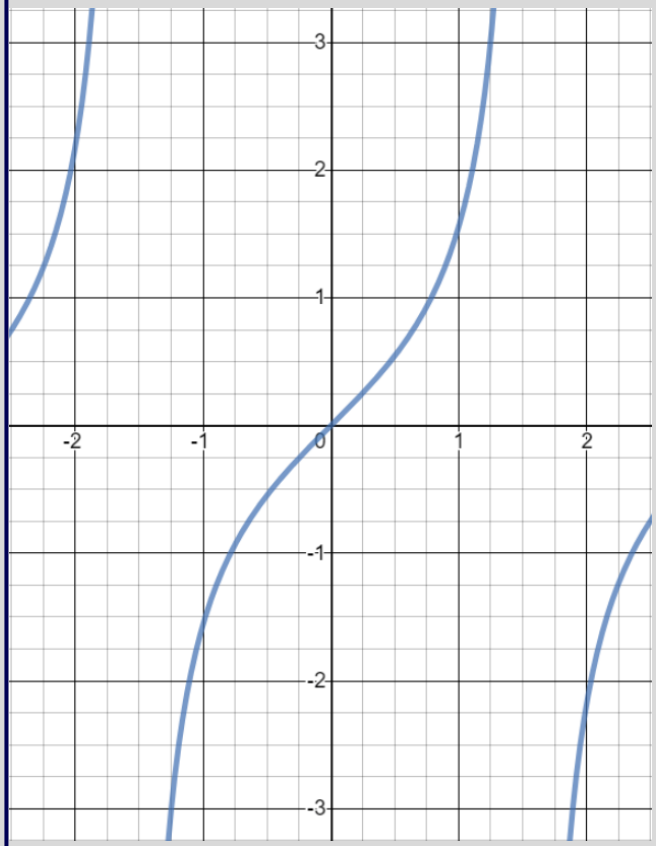
- For $y = \cos(x)$:

$$\frac{d}{dx}(\cos(x)) = -\sin(x)$$

Note that in order for these equations to be valid, the angle x must always be measured in **Radians**.

Derivatives of Trigonometric functions

Example: Find the derivative $\frac{dy}{dx}$ of the function $\mathrm{f(x)} = \tan(x) = \frac{\sin(x)}{\cos(x)}$.



We invoke the quotient rule

$$\frac{\boldsymbol{d}}{\boldsymbol{dx}}\left[\frac{\boldsymbol{f(x)}}{\boldsymbol{g(x)}}\right] = \frac{\boldsymbol{g(x)}\frac{\boldsymbol{d}}{\boldsymbol{dx}}[\boldsymbol{f(x)}] - \boldsymbol{f(x)}\frac{\boldsymbol{d}}{\boldsymbol{dx}}[\boldsymbol{g(x)}]}{[\boldsymbol{g(x)}]^{\boldsymbol{2}}}$$

$$\rightarrow \frac{d}{dx}\left(\frac{\sin(x)}{\cos(x)}\right)$$

$$= \frac{\cos(x)\frac{d}{dx}(\sin(x)) - \sin(x)\frac{d}{dx}(\cos(x))}{\cos(x)^2}$$

$$= \frac{\left(\cos(x) * \cos(x) - \sin(x)\,(-\sin(x))\right)}{\cos(x)^2}$$

$$= \frac{(\cos(x)^2 + \sin(x)^2)}{\cos(x)^2} = \frac{\mathbf{1}}{\mathbf{cos}(\boldsymbol{x})^{\mathbf{2}}} = \boldsymbol{sec(x)^2}$$

Derivatives of Trigonometric functions - Identities

- The following are derivatives for common trigonometric identities:

1. $\frac{d}{dx}(\sin(x)) = \cos(x)$

2. $\frac{d}{dx}(\cos(x)) = -\sin(x)$

3. $\frac{d}{dx}\left(\frac{\sin(x)}{\cos(x)}\right) = \frac{d}{dx}(\tan(x)) = \sec(x)^2$

4. $\frac{d}{dx}\left(\frac{1}{\sin(x)}\right) = \frac{d}{dx}\csc(x) = -\csc(x) * \cot(x)$

5. $\frac{d}{dx}\left(\frac{1}{\cos(x)}\right) = \frac{d}{dx}\sec(x) = \sec(x) * \tan(x)$

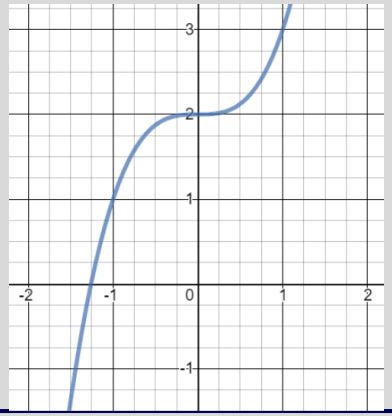
6. $\frac{d}{dx}\left(\frac{\cos(x)}{\sin(x)}\right) = \frac{d}{dx}\cot(x) = -\csc(x)^2$

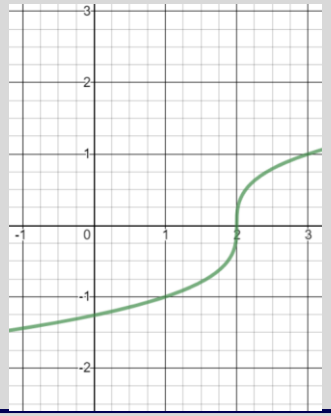
Derivatives of Inverse Trigonometric Function - Introduction

- <u>Inverse functions</u> (written as f^{-1}) are functions that satisfy the following condition for any function $f(x)$:

$$\textbf{If } \boldsymbol{f(x) = y} \textbf{ then the inverse function } \boldsymbol{f^{-1}(y) = x}$$

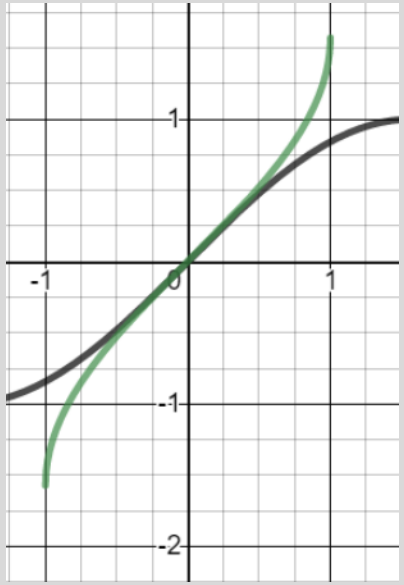
- In other words, the inverse function f^{-1} can be obtained by solving for x in terms of y from the function f.
  - <u>Example</u>: if $f(x) = y = x^3 + 2$ then $f^{-1}(y) = x = \sqrt[3]{y-2}$





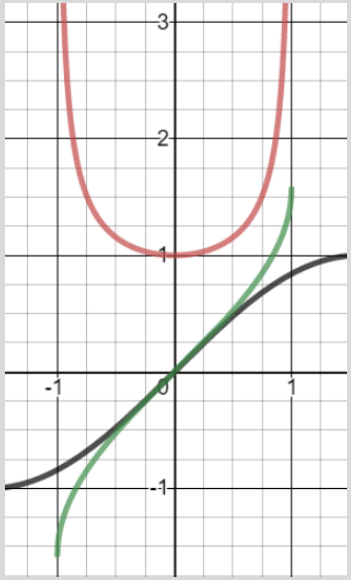
Derivatives of Inverse Trigonometric Function - Introduction

- In trigonometry, the inverse function is useful in finding the angle x if we know the sine or cosine value of x.
  - <u>Example:</u> If $\cos(x) = \frac{1}{2}$, then $x = \cos^{-1}\frac{1}{2} = \frac{\pi}{3}$ (Note: $\cos^{-1} x$ is also called arc-cosine)
- Inverse Trigonometric functions only exist within a defined range, usually for any y-value that is: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.



Derivatives of Inverse Trigonometric Function - Introduction

- We can obtain the derivative $\frac{dy}{dx}$ of inverse trigonometric functions by invoking **<u>Implicit Differentiation</u>** as follows:



$$y = \sin^{-1}(x)$$

$$\rightarrow x = \sin(y)$$

$$\rightarrow \frac{dx}{dx} = \frac{dy}{dx}\frac{d}{dy}(\sin(y)) \leftrightarrow 1 = \frac{dy}{dx} * \cos(y)$$

$$\rightarrow \frac{dy}{dx} = \frac{1}{\cos(y)}$$

- Recall that $\cos(y) = \sqrt{1 - \sin^2(y)}$ (for any y between -1 and 1), hence:

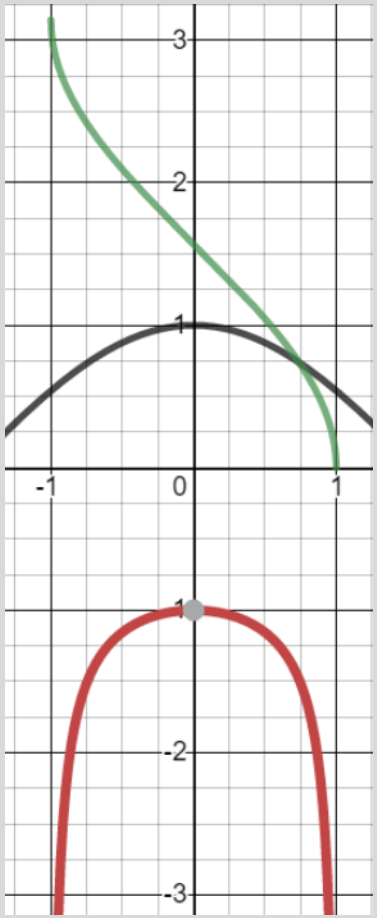
$$\boldsymbol{\frac{d}{dx} sin^{-1}(x) = \frac{dy}{dx}} = \frac{1}{\sqrt{1 - \sin^2(y)}} = \frac{1}{\sqrt{1 - \sin^2(\sin^{-1}(x))}} = \boldsymbol{\frac{1}{\sqrt{1 - x^2}}}$$

Derivatives of Inverse Trigonometric Function - Introduction

- Following a similar process, we can obtain the derivative of the inverse cosine function $f(x) = \cos^{-1} x$ and $f(x) = \tan^{-1} x$

$$\frac{d}{dx} cos^{-1}(x) = -\frac{1}{\sqrt{1 - x^2}}$$

$$\frac{d}{dx} tan^{-1}(x) = \frac{1}{1 + x^2}$$



Note: All of these derivatives are only valid for the case of x

between -1 and 1.

Derivatives of Exponential Functions - Introduction

- Recall the definition of an exponential function:

$$\boldsymbol{f(x) = b^x}$$

- The derivative of $f(x) = e^x$ is:

$$\boldsymbol{\frac{d}{dx} f(x) = \frac{d}{dx} e^x = e^x}$$

Derivatives of Exponential Functions - Introduction

- Recall the identity:

$$\boldsymbol{e^{\ln(a)} = a}$$

- Hence, we can rewrite the 1st equation to be in terms of e:

$$f(x) = b^x = \left(e^{\ln(b)}\right)^x$$

$$\rightarrow \frac{d}{dx} f(x) = \frac{d}{dx} b^x = \frac{d}{dx}\left(e^{\ln(b)}\right)^x = \frac{d}{dx}\left(e^{\ln(b)*x}\right)$$

- Invoking Chain Rule, with $u = \ln(b) * x$, we will find that:

$$\frac{d}{dx}\left(e^{\ln(b)*x}\right) = \frac{d}{du} e^u * \frac{d}{dx}(\ln(b) * x) = e^u * \ln(b) = e^{\ln(b)*x} * \ln(b)$$

Derivatives of Exponential Functions - Introduction

- Thus, we can obtain the general case:

$$\frac{d}{dx} f(x) = \frac{d}{dx} b^x = \ln(b)\, e^{\ln(b) * x} = \ln(b) * b^x$$

Derivatives of Logarithm Functions - Introduction

- Recall the definition of a Natural Log:

$$y = \ln(x) \leftrightarrow \log_e x = y$$

$$\rightarrow e^y = x$$

- We can take the Implicit Derivative of the equation above as follows (while invoking the derivative of an exponential function and chain rule):

$$\ln(e) * e^y \frac{dy}{dx} = \frac{dx}{dx} = 1$$

$$\rightarrow \boldsymbol{\frac{dy}{dx} = \frac{d}{dx} ln(x) = \frac{1}{e^y} = \frac{1}{e^{\ln(x)}} = \frac{1}{x}}$$

Derivatives of Logarithm Functions - Introduction

- For a more general case of log base b:

$$y = \log_b x \rightarrow x = b^y$$

- Then we can take the natural log of both sides of the equation above:

$$\ln(x) = \ln(b^y)$$

$$\leftrightarrow \ln(x) = y\ln(b)$$

$$\leftrightarrow y = \frac{\ln(x)}{\ln(b)} = \log_b(x)$$

Derivatives of Logarithm Functions - Introduction

- Next, we shall take the derivative of y with respect to x:

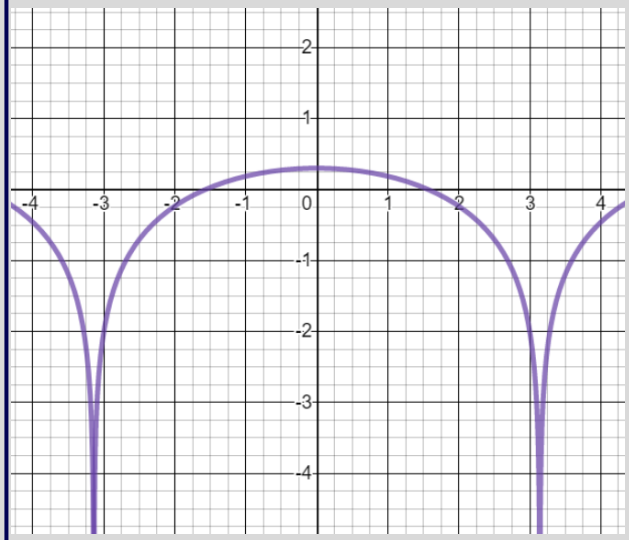
$$y = \frac{\ln(x)}{\ln(b)} = \log_b(x)$$

$$\rightarrow \frac{dy}{dx} = \frac{d}{dx}\log_b(x) = \frac{d}{dx}\left(\frac{\ln(x)}{\ln(b)}\right) = \frac{1}{\ln(b)}\frac{d}{dx}\ln(x)$$

$$\rightarrow \boldsymbol{\frac{d}{dx}\log_b(x) = \frac{1}{\ln(b) * x}}$$

Derivatives of Logarithm Functions - Example

Example: Find the derivative $\frac{dy}{dx}$ of the function $\mathrm{f(x)} = \log_{10}(1 + \cos x)$.



Note: There is a discontinuity at value $x = \pi$

(1) Note that we have 2 functions: $u = 1 + \cos(x)$ and $y = \log_{10}(u)$. Thus we must invoke Chain rule.

$$\frac{dy}{dx} = \frac{d}{du}(\log_{10}(u)) * \frac{d}{dx}(1 + \cos(x))$$

(2) Recall that:

$$\mathbf{\frac{d}{dx}\log_b(x) = \frac{1}{\ln(b) * x}}$$

Thus:

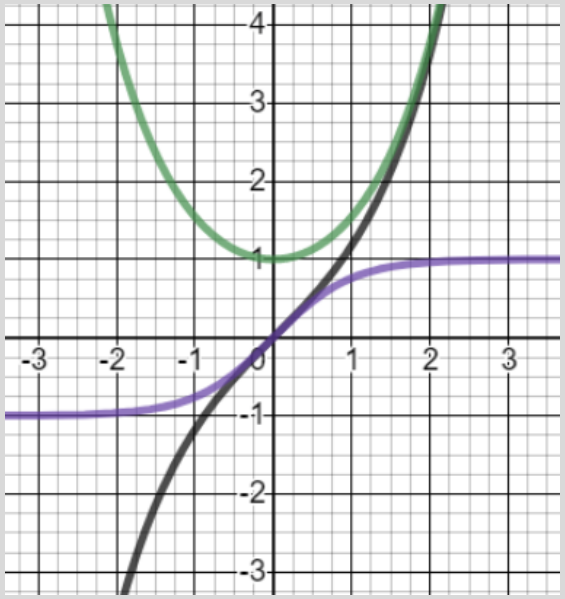
$$\frac{d}{du}(\log_{10}(u)) = \frac{1}{\ln(10) * u} = \frac{1}{\ln(10) * (1 + \cos(x))}$$

(3) Hence we have the final result:

$$\frac{dy}{dx} = \frac{1}{\ln(10) * (1 + \cos(x))}(-\sin(x))$$

Derivatives of Hyperbolic Functions - Introduction

- The hyperbolic functions are based on the unit hyperbola ($x^2 - y^2 = 1$), which could be defined in terms of exponential functions:



$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$$

Derivatives of Hyperbolic Functions - Introduction

- Invoking the Rules we have learned thus far, it is simple to find the derivatives of each function:

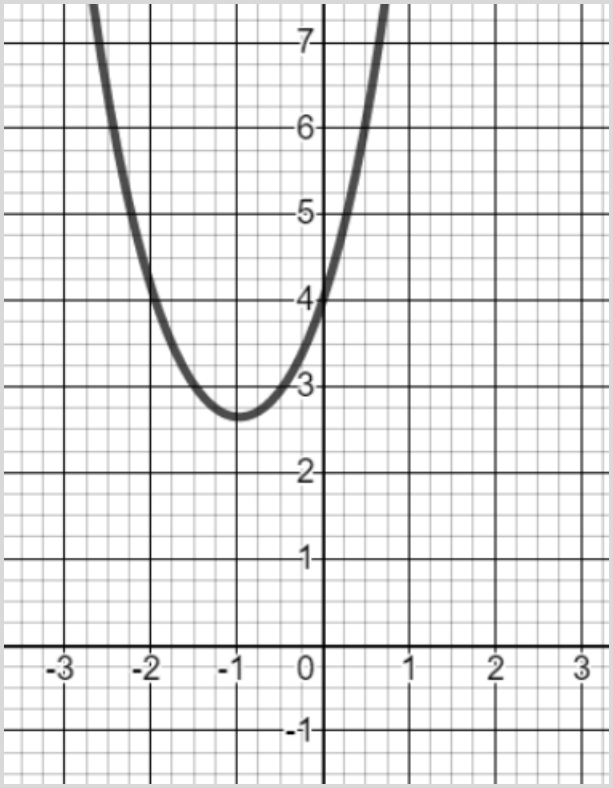
$$\frac{d}{dx}\sinh(x) = \frac{e^{x} + e^{-x}}{2} = \cosh(x)$$

$$\frac{d}{dx}\cosh(x) = \frac{e^{x} - e^{-x}}{2} = \sinh(x)$$

$$\frac{d}{dx}\tanh(x) = \frac{1}{\cosh^{2} x}$$

Derivatives of Hyperbolic Functions - Example

Example: Find the derivative $\frac{dy}{dx}$ of the function $\mathrm{f(x)} = 3\sinh(x) + 4\cosh(x)$.



We invoke the formula for finding derivatives of Hyperbolic Functions:

$$\frac{d}{dx}f(x) = \frac{d}{dx}(3\sinh(x)) + \frac{d}{dx}(4\cosh(x))$$
$$= \mathbf{3\cosh(x) + 4\sinh(x)}$$

Differentiation Formulas – Exercises

Find the derivatives $\frac{dy}{dx}$ for the following functions:

1. $f(x) = 6\sin\left(\frac{2}{x}\right)$

2. $f(x) = x^5 e^{3x}$

3. $f(x) = \ln\left(\frac{4x+9}{4x-9}\right)$

4. $f(x) = \tan^{-1}\left(\frac{6x+5}{4}\right)$

Applications of Differentiation

How changing one thing affect the changing of another.

RMIT UNIVERSITY

Applications of Differentiation– Main topics

1. **Small Change Approximations**
2. **Related Rate**
3. **Optimization problems.**
4. **Finding Normal line to a curve**
5. **Finding displacement, velocity, and acceleration of moving objects.**

Applications of Differentiation– Small Change Approximations

- Recall the Limit definition of a Derivative:

$$f'(x) = \lim_{h \to 0} \left(\frac{f(x+h) - f(x)}{h} \right)$$

- For small values of h, we can obtain the approximation:

$$f'(x) = \frac{f(x+h) - f(x)}{h} \rightarrow \boldsymbol{f(x+h) = f'(x) * h + f(x)}$$

- In this case, h is a very small change to the x-value.

Applications of Differentiation– Small Change Approximations Example

Example: A car on the highway ran past 2 policemen standing 20 meters apart. Camera recording indicates that it took the car 0.8 seconds to travel from the 1st policeman to the 2nd policeman. If the speed limit is 60 km/hr at this location, is the car speeding?

Answer:

(1) We invoke the Small Change Approximation formula:

$$\boldsymbol{f(x+h) = f'(x) * h + f(x)}$$

In the case of the car, distance is a function of time, so x is time. The change in time is $h = 0.8\ second$. We are looking for $f'(x)$.

(2) Thus we have the speed of the car:

$$f'(x) = \frac{f(x+h) - f(x)}{h} = \frac{20}{0.8} = 25\ meters/second$$

Finally, we convert the speed of the car to km/hr:

$25\frac{meters}{second} * 3600\frac{seconds}{hour} * 0.001\frac{km}{m} = 90\ km/hr$→Yes, the car is speeding.

Applications of Differentiation– Related Rate

- Related Rate problems are where the rate of change of quantity A is changing relative to the rate of change of quantity B.
- Mathematically, the problem has the form:

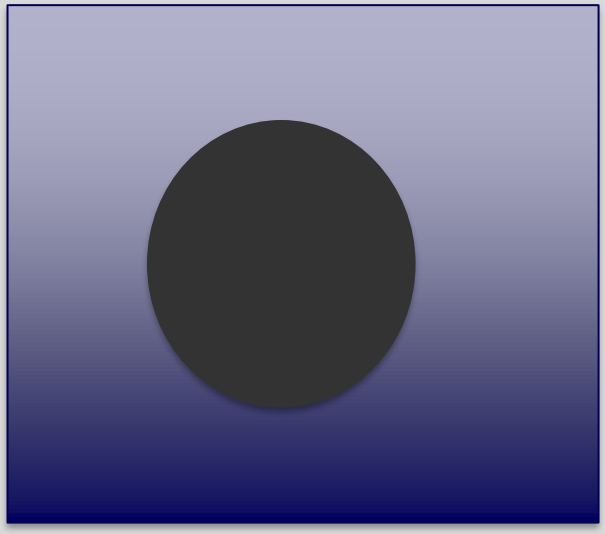
$$y = y(u) \text{ with } u = u(x)$$

$$\text{Thus } \frac{dy}{du} = \frac{dy}{du} \cdot \frac{du}{dx}$$

- To solve Related problems, follow these steps:
  1. Draw a diagram
  2. Express the given information in terms of derivatives.
  3. Write and equation that relates the various quantities of the problem.
  4. Use the Chain rule to differentiate both sides.
  5. Substitute the given information and solve for the unknown rate.

Applications of Differentiation–
Related Rate Example

<u>Example</u>: In the middle of the ocean there is a perfectly circular oil slick. Scientists observed that the oil slick's radius is increasing by 5 m/s in all directions. Assuming the oil slick remains circular, determine the rate at which its area is increasing when the oil slick is 20 meters wide.



Answer:

(1) The area of a circle is given by:

$$A = A(r) = \pi r^2$$

(2) In our example, r is a function of time since r is changing, hence

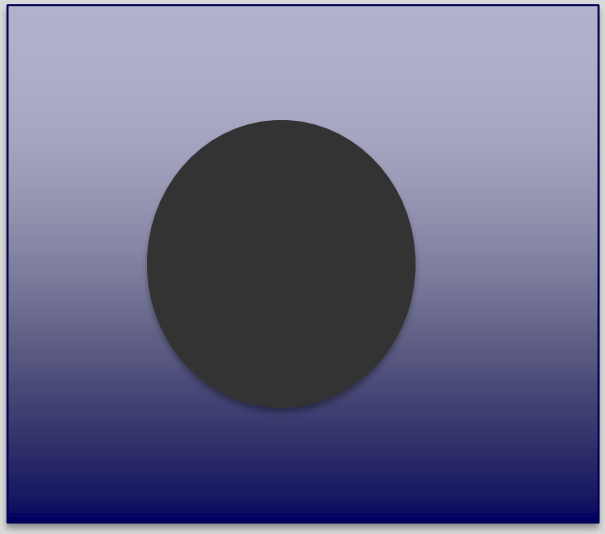
$$\frac{dr}{dt} = 5\ m/s$$

(3) We take the derivative of the area with respect to time (with Chain Rule):

$$\frac{d}{dt}A = \frac{d}{dt}(\pi r^2) = 2\pi r * \frac{dr}{dt}$$

Applications of Differentiation–
Related Rate Example

<u>Example</u>: In the middle of the ocean there is a perfectly circular oil slick. Scientists observed that the oil slick's radius is increasing by 5 m/s in all directions. Assuming the oil slick remains circular, determine the rate at which its area is increasing when the oil slick is 20 meters wide.



Answer:

(3) We take the derivative of the area with respect to time (with Chain Rule):

$$\frac{d}{dt}A = \frac{d}{dt}(\pi r^2) = 2\pi r * \frac{dr}{dt}$$

(4) We substituting the quantities which we know into the equation, when $r(t) = 10\ meters$

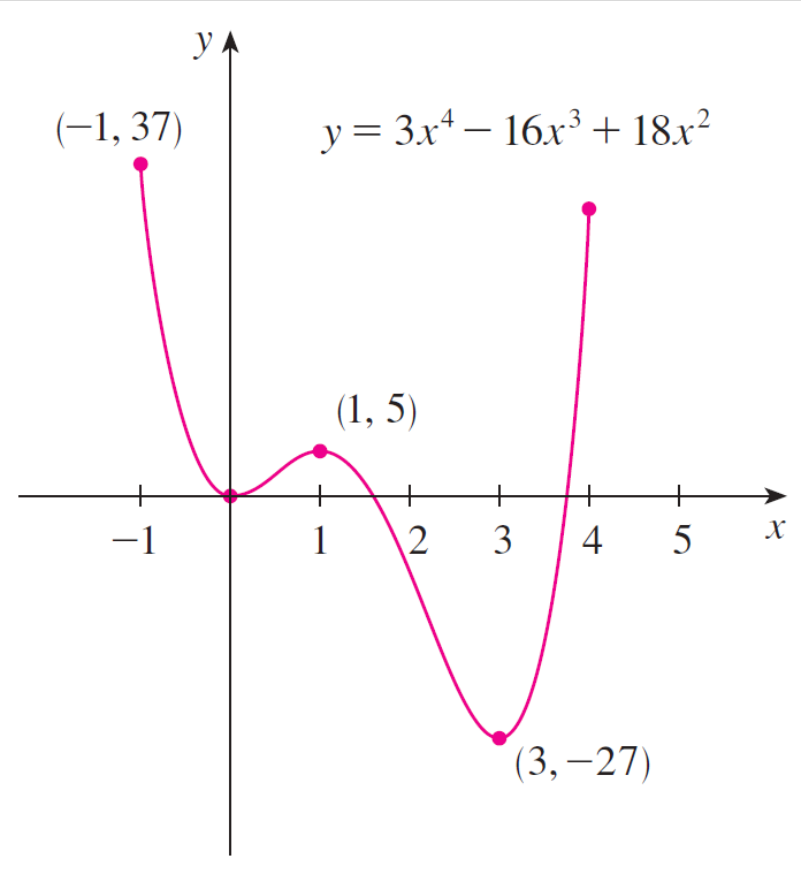
$$\frac{d}{dt}A = 2 * \pi * 10 * 5 = 100\pi \approx \mathbf{314\ m^2/s}$$

Applications of Differentiation– Optimization

- Optimization is where we find the **optimal (best) way** to do something.
- Optimization is achieved by finding the extreme values of a function, which are its maximum and minimum value.
- The maximum and minimum values also belong to the Stationary points set (which includes max points, min points, and inflection points)
- A graph may have many local maximums and minimums but it should only have one absolute maximum and/or one absolute minimum.

Applications of Differentiation– Optimization

- Given the curve to the function: $y = 3x^4 - 16x^3 + 18x^2$



Observe:

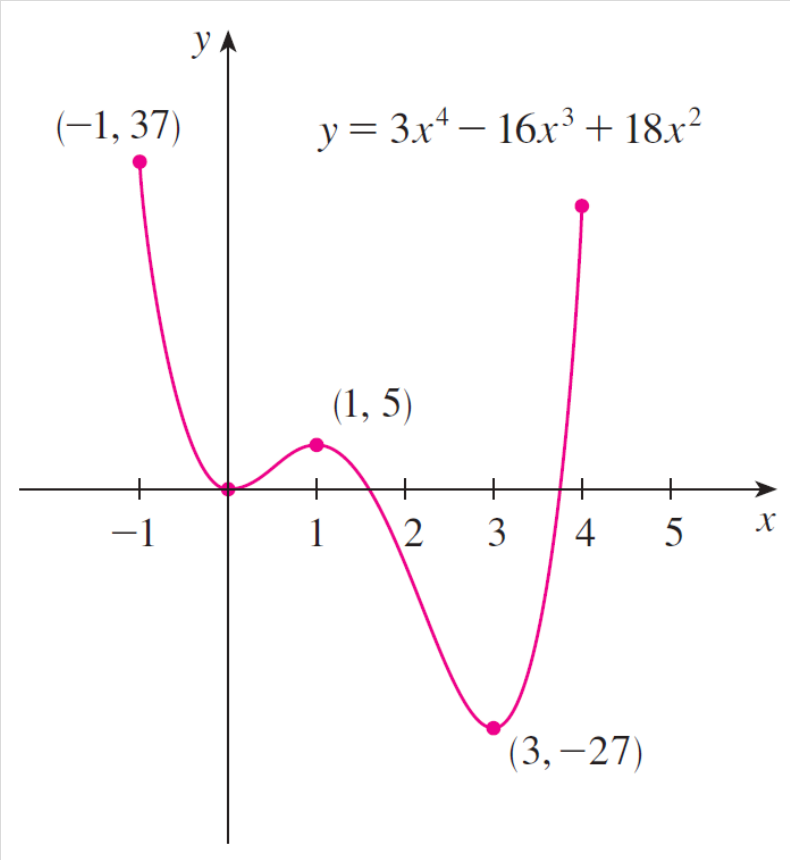
(1) The curve has a local Minimum at point (3, -27).
(2) The curve has a local Maximum at point (1, 5).
(3) The curve has a local Minimum at point (0, 0).

Applications of Differentiation– Optimization – Max, Min

- Mathematically, the local maximum and minimum points for any function $\boldsymbol{y = f(x)}$ within a domain $\boldsymbol{a \leq x \leq b}$ are:
  - **$\boldsymbol{f(a)}$ is a local maximum value at $\boldsymbol{a}$ if:**
    - $\boldsymbol{f'(a) = 0 \text{ and } f''(a) < 0}$
  - **$\boldsymbol{f(a)}$ is a local minimum value at $\boldsymbol{a}$ if:**
    - $\boldsymbol{f'(a) = 0 \; and \; f''(a) > 0}$
  - **$\boldsymbol{f(a)}$ is an inflection point at $\boldsymbol{a}$ if:**
    - $\boldsymbol{f'(a) = 0 \; and \; f''(a) = 0}$

Applications of Differentiation– Optimization – Max, Min examples

Example: Consider the function $y = 3x^4 - 16x^3 + 18x^2$. Using the derivative y', find all the maximum and minimum points of this function.



(1) Invoking the derivative function y':

$$y' = 12x^3 - 48x^2 + 36x$$
$$= 12x(x^2 - 4x + 3)$$
$$= 12x(x-1)(x-3) = 0$$

Hence the roots of the function $f'(x)$ are $x_1 = 0, x_2 = 1, x_3 = 3$

(2) Find the corresponding y-value for each x-value using the original function:

$f(x_1) = 3(0)^4 - 16(0) + 18(0)^2 = 0$

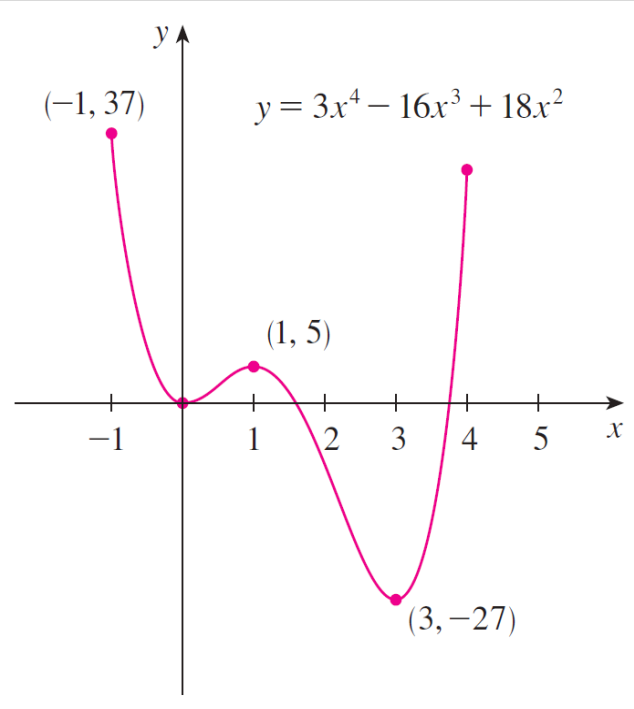
$f(x_2) = 3(1)^4 - 16(1) + 18(1)^2 = 5$

$f(x_1) = 3(3)^4 - 16(3) + 18(3)^2 = -27$

Hence we have 3 extreme values $P_1(0,0), P_2(1,5), P_3(3,-27)$

Applications of Differentiation– Optimization – Max, Min examples

Example: Consider the function $y = 3x^4 - 16x^3 + 18x^2$. Using the derivative y', find all the maximum and minimum points of this function.



Check the graphs of y' and y'' here.

(3) To determine if those points are maximum or minimum values, we look at the function y'', which is:

$$y'' = \frac{d}{dx}y' = 36x^2 - 96x + 36$$

Recall that if an extreme value has $f''(x) < 0$ then it is a maximum value and if $f''(x) > 0$, it is a minimum value.

$f''(x_1) = f_1''(0) = 36(0)^2 - 96(0) + 36 = 36 > 0$
$f''(x_2) = f_2''(1) = 36(1)^2 - 96(1) + 36 = -24 < 0$
$f''(x_1) = f_3''(3) = 36(3)^2 - 96(3) + 36 = 72 > 0$

(4) Hence $P_1(0,0)$ is a minimum value, $\boldsymbol{P_2(1,5)}$ **is a maximum** value, and $\boldsymbol{P_3(3,-27)}$ is a **minimum** value.

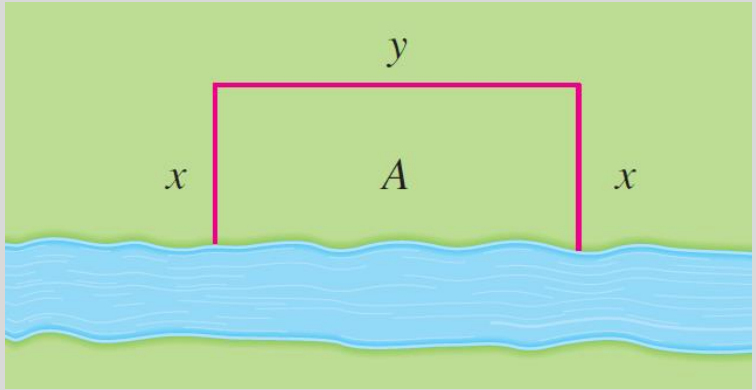
Applications of Differentiation– Optimization – Solving Procedures

To solve Optimization Problems, follow these steps:

1. Read the problem and identify the unknowns and given quantities/condition.
2. Draw a diagram of the problem.
3. Find an equation to relate the unknown (that has to be maximized or minimized) with the given information.
4. Use the method discussed to find the **absolute** maximum or **absolute** minimum points of the function from step 3.

Applications of Differentiation– Optimization examples

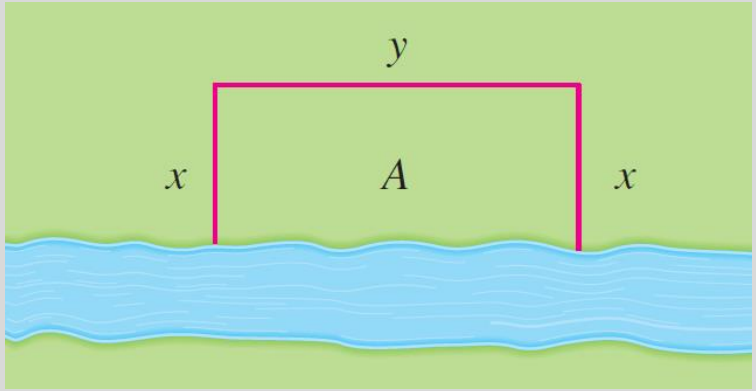
Example: A farmer has 2400 meters of fencing material and he wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?



(1) Identify the unknown: we are interested in the width x and length y of the rectangular area where the area is the largest.
(2) Draw the diagram.
(3) Find the equation to relate the unknown to the known: We can form the following equations from the given:
   (a) Shape of the area is rectangular and the total number of fencing is limited so:
   $$2x + y = 2400$$
   (b) The area of the fenced area is:
   $$x * y = A$$
   (c) We are interested in finding when A is maximum.

Applications of Differentiation– Optimization examples

Example: A farmer has 2400 meters of fencing material and he wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?



(4) Find the absolute maximum value of equation 3(b):

$$A = xy$$

From equation 3(a), we have:

$$2x + y = 2400$$

$$\rightarrow y = 2400 - 2x = 2(1200 - x)$$

Substitute the quantity above into y:

$$A = x\big(2(1200 - x)\big) = 2400x - 2x^2$$

Hence, we can find the rate of change of the area with respect to the width x as:

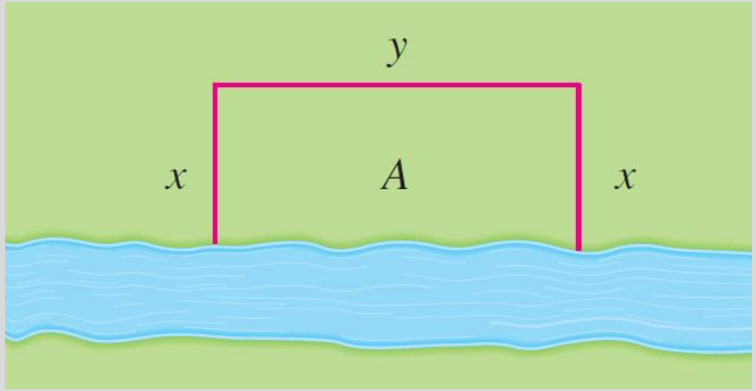
$$\frac{d}{dx}A = \frac{d}{dx}(2400x) - \frac{d}{dx}(2x^2) = 2400 - 4x$$

Hence, we can find the root of the equation above:

$$\frac{d}{dx}A = 0 = 2400 - 4x \rightarrow x = \frac{2400}{4} = 600 \text{ meters}$$

Applications of Differentiation– Optimization examples

Example: A farmer has 2400 meters of fencing material and he wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?



(4) (cont.) Thus we have one extreme value at $x = 600$, the corresponding value of y is:

$$y = 2400 - 2(600) = 1200 \text{ meters}$$

To determine if this is a maximum or minimum point, we take the 2nd derivative:

$$\frac{d}{dx}A' = \frac{d}{dx}2400 - \frac{d}{dx}(4) = -4 < 0$$

Hence, this is the maximum point.

(5) Finally we can conclude that the largest area possible is **600 meters x 1,200 meters= 720,000 m^2.**

Applications of Differentiation– Finding a Normal Line to a curve

- In Optical Engineering, it is useful to know the equation of a line that is normal to any given point on a curve.
  - Note: "Normal" means it forms a 90º angle with the line.

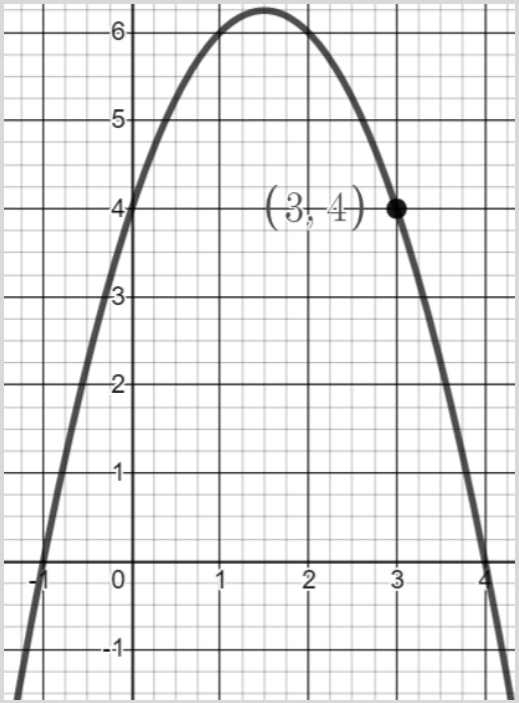
- By definition, given a line A: $f_1(x) = m_1 x + b_1$, the normal line to line A will be line B: $f_2(x) = m_2 x + b_2$ where $m_2 = -\frac{1}{m_1}$

- The slope of the normal line to the curve at any point is

$$\boldsymbol{m = -\frac{1}{f'(a)}}$$

Applications of Differentiation– Optimization examples

Example: A new camera lenses design has a cross section that can be approximated with the equation $y = -x^2 + 3x + 4$. Find the line equation of the normal line to this lenses at point $P(3, 4)$.



(1) Find the derivative function of y:

$$y = -x^2 + 3x + 4$$
$$\rightarrow y' = -2x + 3$$

(2) Find the slope of the tangent line at this point:

$$f'(3) = -2 * 3 + 3 = -3$$

(3) Find the slope of the normal line at this point:

$$m_{normal} = -\frac{1}{f'(3)} = -\frac{1}{-3} = \frac{1}{3}$$

(4) Find the line equation of the normal line:

$$y = m(x - x_0) + y_0$$
$$\rightarrow y = \frac{1}{3}(x - 3) + 4$$
$$\rightarrow \boldsymbol{y = \frac{1}{3}x + 3}$$

Illustration here.

Applications of Differentiation– Displacement, velocity, and acceleration

- The displacement (or position) function of an object is given as a <u>parametric curve</u> with the time <u>t as the variable</u>.
- Thus given the displacement function of an object as:

$$\boldsymbol{d = f(t)}$$

The velocity function of that object will be the derivative of the displacement function:

$$\boldsymbol{v(t) = d'(t)}$$

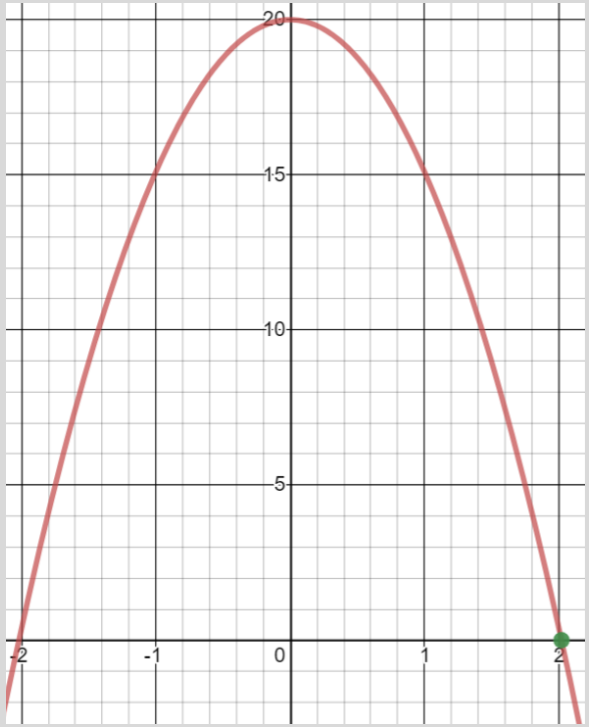
The acceleration function of that object will be the derivative of the velocity function:

$$\boldsymbol{a(t) = v'(t) = d''(t)}$$

<u>Extra:</u> The derivative of the acceleration function will yield a quantity called “jerk” which is the sudden shift in acceleration.

Applications of Differentiation– Optimization examples

Example: A ball is being dropped from the top of a building. Its vertical height above the ground (where the ground is at 0 meter) is given by the equation: $f(t) = y = -4.9t^2 + 20$. Find: a) How long does it takes for the ball to reach the ground, b) The ball's velocity 1 second after it was dropped, c) What is its acceleration?



(1) Find the velocity and acceleration functions:

$$f(t) = -4.9t^2 + 20$$
$$f'(t) = -9.8t$$
$$f''(t) = -9.8$$

(2) Observe that its acceleration function is a constant and does not depend on time so the balls' acceleration is always $\boldsymbol{-9.8\ m/s^2}$

(3) To find how long does it take the ball to reach the ground, we find the roots of the equation:

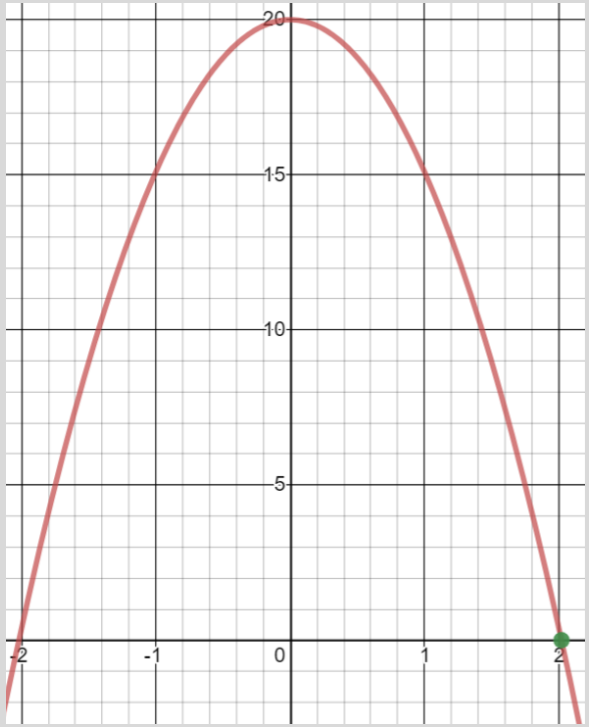
$$f(t) = 0 = -4.9t^2 + 20$$

$$\rightarrow t = \pm\sqrt{\frac{20}{4.9}} = \pm 2.02\ seconds.$$

Since we start time at 0, therefore, the ball will reach the ground in **2.02 seconds**.

Applications of Differentiation– Optimization examples

<u>Example:</u> A ball is being dropped from the top of a building. Its vertical height above the ground (where the ground is at 0 meter) is given by the equation: $f(t) = y = -4.9t^2 + 20$. Find: a) How long does it takes for the ball to reach the ground, b) The ball's velocity 1 second after it was dropped, c) What is its acceleration?



(4) To find the ball's velocity 1 second after it was dropped, we use the velocity equation:

$$f'(1) = -9.8(1) = \boldsymbol{-9.8\ m/s}$$

Hence the ball was travelling downward at a speed of 9.8 m/s 1 second after it was dropped.

Questions?

The only stupid question is the one that was not asked.

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